

6.7960

Deep

Learning



Instructor Phillip Isola
phillipi at mit dot edu
OH: TBD



Instructor Kaiming He
kaiming at mit dot edu
OH: TBD



Course Assistant
Taylor Braun
tvbraun at mit dot edu



Head TA Ning Zhang
ningrz at mit dot edu
OH: TBD



TA Amit Schechter
amitsch at mit dot edu
OH: TBD



TA Aryan Bora
abora at mit dot edu
OH: TBD



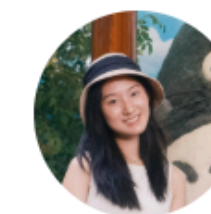
TA David Baek
dbaek at mit dot edu
OH: TBD



TA Maggie Lin
maggiejl at mit dot edu
OH: TBD



TA Suraj Reddy
surajrdy at mit dot edu
OH: TBD



Head TA Lana Xu
ylanaxu at mit dot edu
OH: TBD



TA Amy Chang
akchang at mit dot edu
OH: TBD



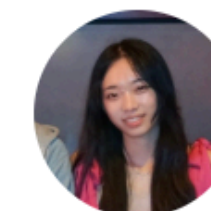
TA Ashkan Soleymani
ashkanso at mit dot edu
OH: TBD



TA Gabriel Manso
gmanso at mit dot edu
OH: TBD



TA Rebecca Wang
rebecca1 at mit dot edu
OH: TBD



TA Tina Wang
txw at mit dot edu
OH: TBD



TA Xingjian Bai
xbai at mit dot edu
OH: TBD



TA Aimee Yu
aimeeyu at mit dot edu
OH: TBD



TA Anakha Ganesh
anakhag at mit dot edu
OH: TBD



TA Cindy Wei
cindywei at mit dot edu
OH: TBD



TA Jake Austin
j_austin at mit dot edu
OH: TBD



TA Shobhita Sundaram
shobhita at mit dot edu
OH: TBD



TA Vanessa Xiao
vzxiao at mit dot edu
OH: TBD

website:

<https://deeplearning6-7960.github.io/>

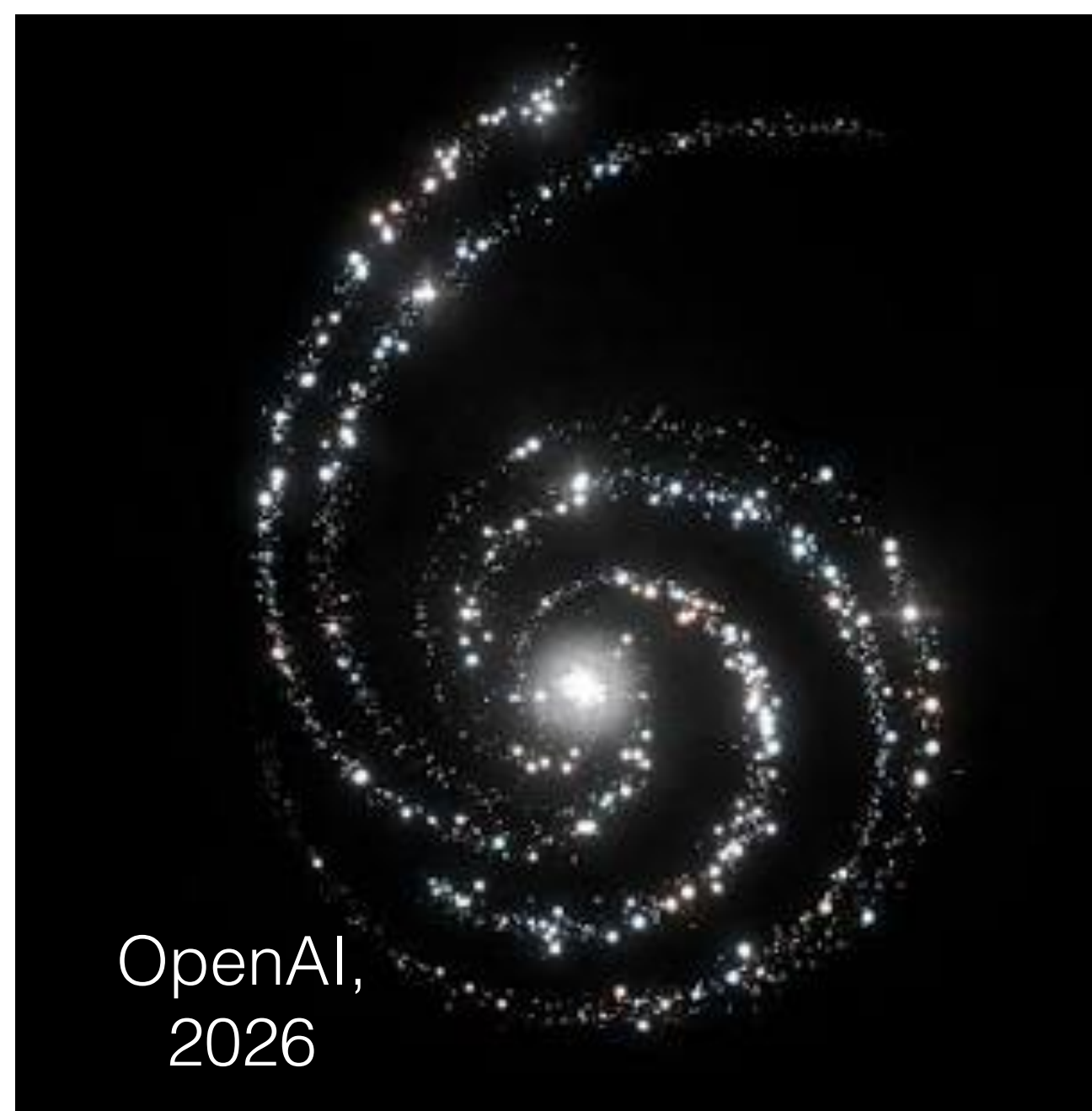
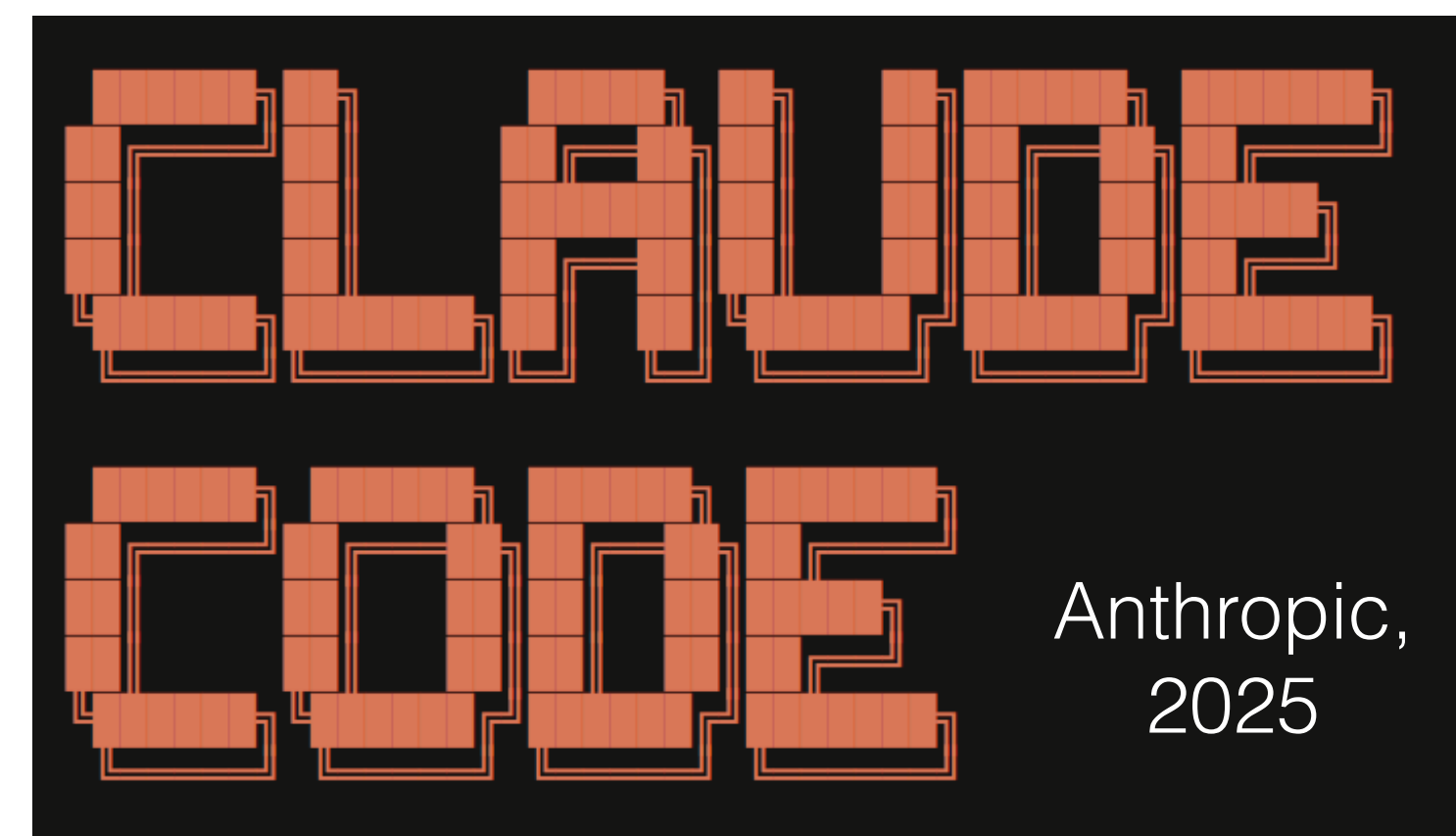


ChatGPT

OpenAI, 2022



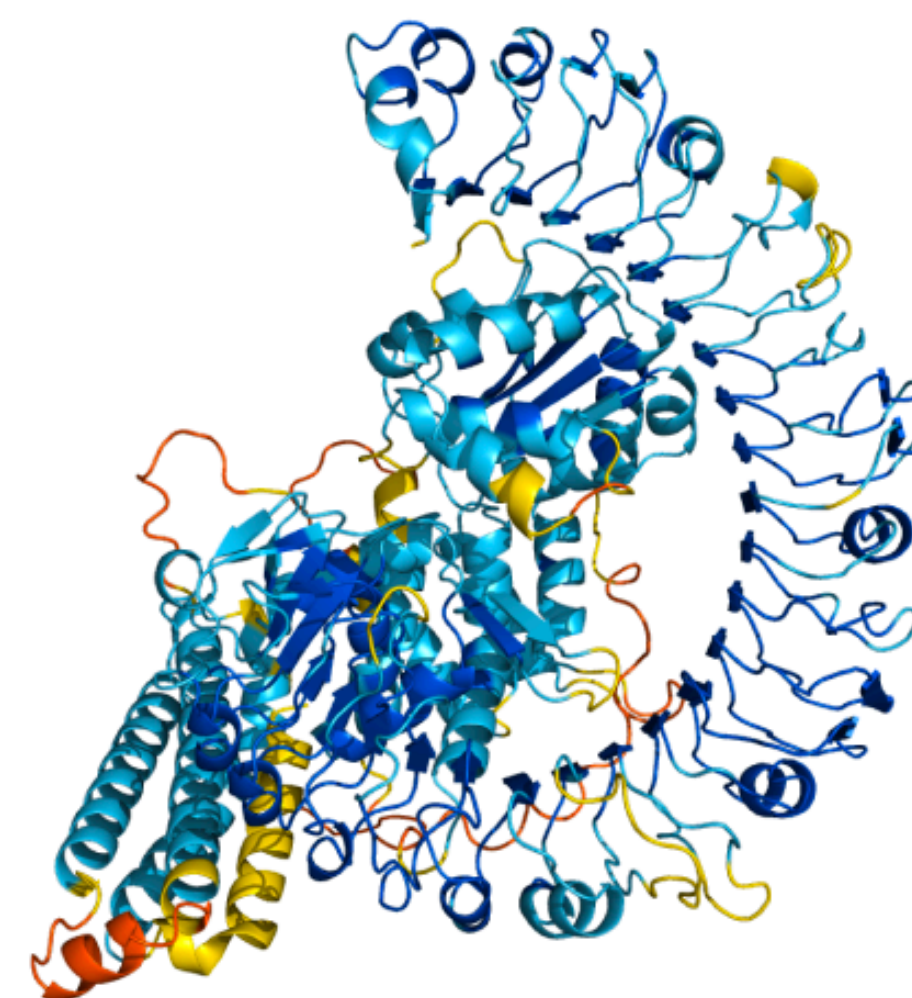
OpenAI, 2025



OpenAI,
2026



Google,
2025



DeepMind, 2021



DeepMind,
2016

What is “deep learning”?

1. **Artificial Neural Nets:** A class of machine learning models built by composing elementary computational components
2. **Differentiable Programming:** A programming paradigm where we parameterize parts of the program and let gradient-based optimization tune the parameters

Course scope

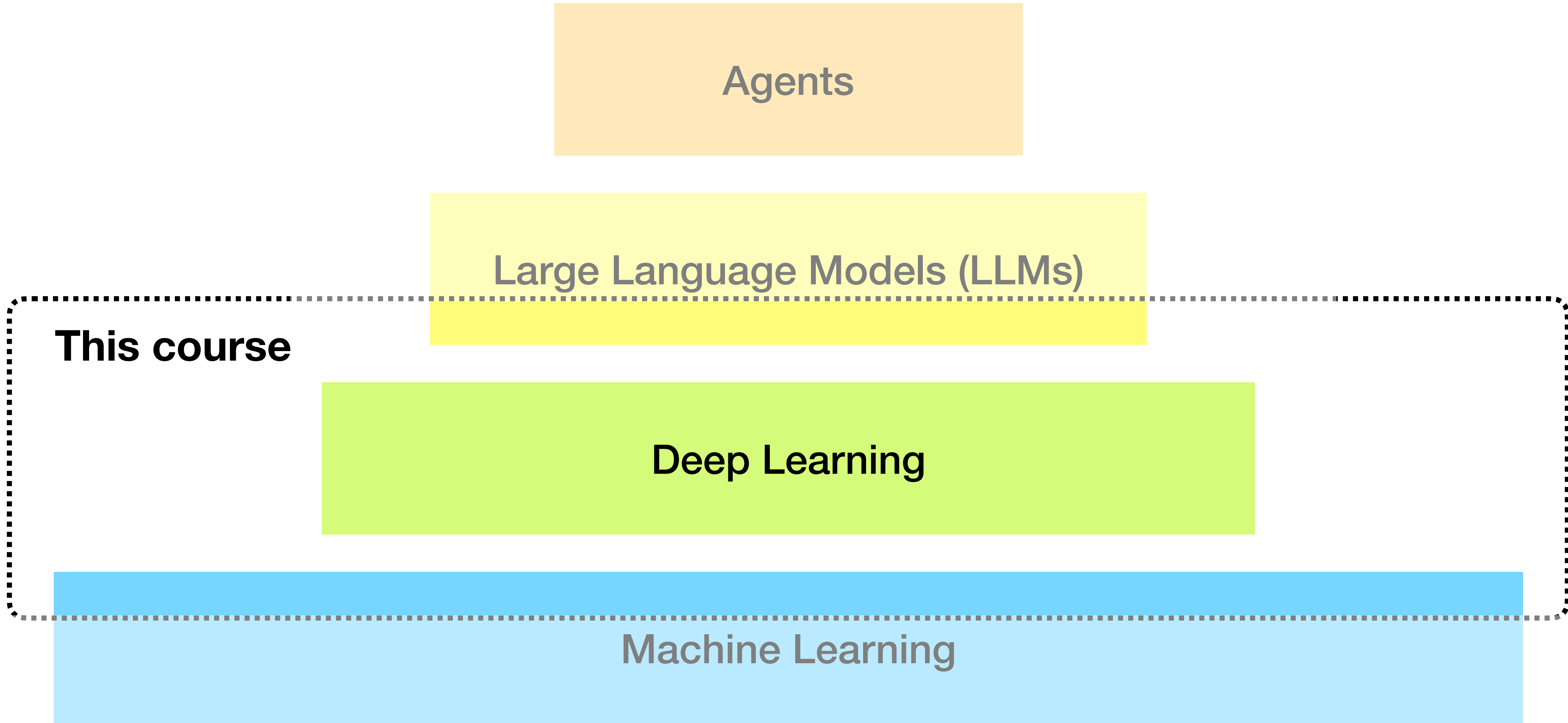
Agents

Large Language Models (LLMs)

This course

Deep Learning

Machine Learning



Logistics

- Lectures: Tuesday, Thursday 1:00 - 2:30 PM in room **45-230**.
- We will be using both Piazza and Canvas for announcements.
 - Refer to: [Piazza](#) (all questions), [Canvas](#) (announcements and homework release), [Gradescope](#) (HW submission and grades).
- All extension requests *must* go through S3 or GradSupport
 - For *any* personal or logistical questions, such as regarding absences, accommodations, etc, please email the course email, not the instructors directly.
 - We provide late days (more later)
 - We cannot support OX requests

Grading

- **Problem sets (20%)**
 - 5 psets, 4% each, each ~2 weeks long
 - Derivations, written responses, and coding
 - Grading will be **pass/fail**
 - **AI agents do grading and give feedback**, with oversight by TAs
- **Midterm Exam (30% + 5% for practice exam) + Final Exam (40% + 5% for practice exam)**
 - 2-hour exam, closed book, no computers, 1 double-sided page of handwritten notes
 - Practice exams will be provided
 - Practice exams will be graded pass/fail and given feedback by AI agents.

Schedule

Week 1				
Thu 9/10	Course overview, introduction to deep learning	Kaiming He		
Week 2				
Tue 9/15	How to train a neural net	Phillip Isola		pset 1 out
Thu 9/17	Approximation theory	Phillip Isola		
Week 3				
Tue 9/22	Architectures: Grids	Kaiming He		
Thu 9/24	Architectures: Memory and Sequence Modeling	Kaiming He		
Week 4				
Tue 9/29	Architectures: Transformers	Phillip Isola		pset 1 due pset 2 out
Thu 10/1	Generalization Theory	Phillip Isola		
Week 5				
Tue 10/6	Going Deep with Neural Networks	Kaiming He		

Thu 10/8	Representation Learning Methods	Kaiming He		
Week 6				
Tue 10/13	No class — Monday schedule			pset 2 due pset 3 out
Thu 10/15	Representation Learning: Weight-Space Geometry	Phillip Isola		
Week 7				
Tue 10/20	Representation Learning: Information Theory	Kaiming He		
Thu 10/22	Foundation models: pre-training	Phillip Isola		
Week 8				
Tue 10/27	Foundation models: scaling laws	Phillip Isola		pset 3 due pset 4 out
Thu 10/29	Midterm			

Week 9				
Tue 11/3	Generative models: basics	Kaiming He		
Thu 11/5	Generative models: VAE and GAN	Kaiming He		
Week 10				
Tue 11/10	Generative models: Diffusion and Flows	Kaiming He		pset 4 due pset 5 out
Thu 11/12	Generalization (OOD)	Phillip Isola		
Week 11				
Tue 11/17	Transfer learning: Models and Data	Phillip Isola		
Thu 11/19	Inference-time Algorithms	Phillip Isola		

Week 12				
Tue 11/24	Guest Lecture 1	TBD		
Thu 11/26	No class — Thanksgiving			
Week 13				
Tue 12/1	Evaluation	Phillip Isola		pset 5 due
Thu 12/3	Applying Deep Learning to Your Problems	Kaiming He		
Week 14				
Tue 12/8	Guest Lecture 2	TBD		
Thu 12/10	Guest Lecture 3	TBD		
Week 15				
Tue 12/15	Final Exam			

Tutorials

- We offer two tutorial topics:
 - **PyTorch**
 - **[NEW] Agentic Coding** (Claude Code / Codex)
- Dates/Places TBD
 - Given by TAs
 - Each topic available at two slots

Weekly Office Hours

- Starting next week (rooms and schedules TBA on website)
- TA office hours for pset questions
- Instructor office hours for general content or research questions

Collaboration Policy

- Psets should be written up individually and should reflect your own individual work. However, you may discuss with your peers, TAs, and instructors.
- You should not copy or share complete solutions or ask others if your answer is correct (in person or via piazza/canvas)
- If you work with anyone on the pset (other than TAs and instructors), list their names at the top of the pset

AI Assistants Policy

- Our policy for using ChatGPT and other AI assistants is *identical* to our policy for using human assistants.
- This is a deep learning class and you *should* try out all the latest AI assistants (they are pretty much all using deep learning). It's very important to play with them to learn what they can do and what they can't do. That's a part of the content of this course.
- Just like you can come to office hours and ask a human questions (about the lecture material, clarifications about pset questions, tips for getting started, etc), you are very welcome to do the same with AI assistants.
- But: just like you are not allowed to ask an expert friend to do your homework for you, you also should not ask an expert AI.
- If it is ever unclear, just imagine the AI as a human and apply the same norm as you would with a human.
- If you work with any AI on a pset, briefly describe which AI and how you used it at the top of the pset (a few sentences are enough).

Late Policy

- Homeworks will not be accepted more than 7 days after the deadline.
- The grade on a homework received n days after the deadline ($n \leq 7$) will be multiplied by $(1 - n/14)$. We will round up to units of full days; submitting 1 hour late counts as using 1 late day.
- Ten penalty days will be automatically waived for each student.
 - Intended to replace special requests for normal circumstances of life — being behind on work, forgetting the deadline, having a conference to attend, etc. **No further extensions will be provided.**
 - *Any* additional extension request (i.e. serious medical issue or life event) must go through S3 or GradSupport, we will work with them directly to find a solution
- No course incomplete will be able to be supported, please prepare accordingly

Keeping track of everything

(All info on the website!)

- Course webpage: <https://deeplearning6-7960.github.io/>
 - High-level course information, schedule
- Canvas: <https://canvas.mit.edu/courses/40032>
 - Syllabus, course recordings, psets
- Piazza: <https://piazza.com/class/mtncdc6cx5o3p9#>
 - Interact with TAs, ask questions
- GradeScope: https://canvas.mit.edu/courses/40032/external_tools/4863
 - Psets are submitted and graded here

Lecture 1: Introduction to Deep Learning

Speaker: Kaiming He

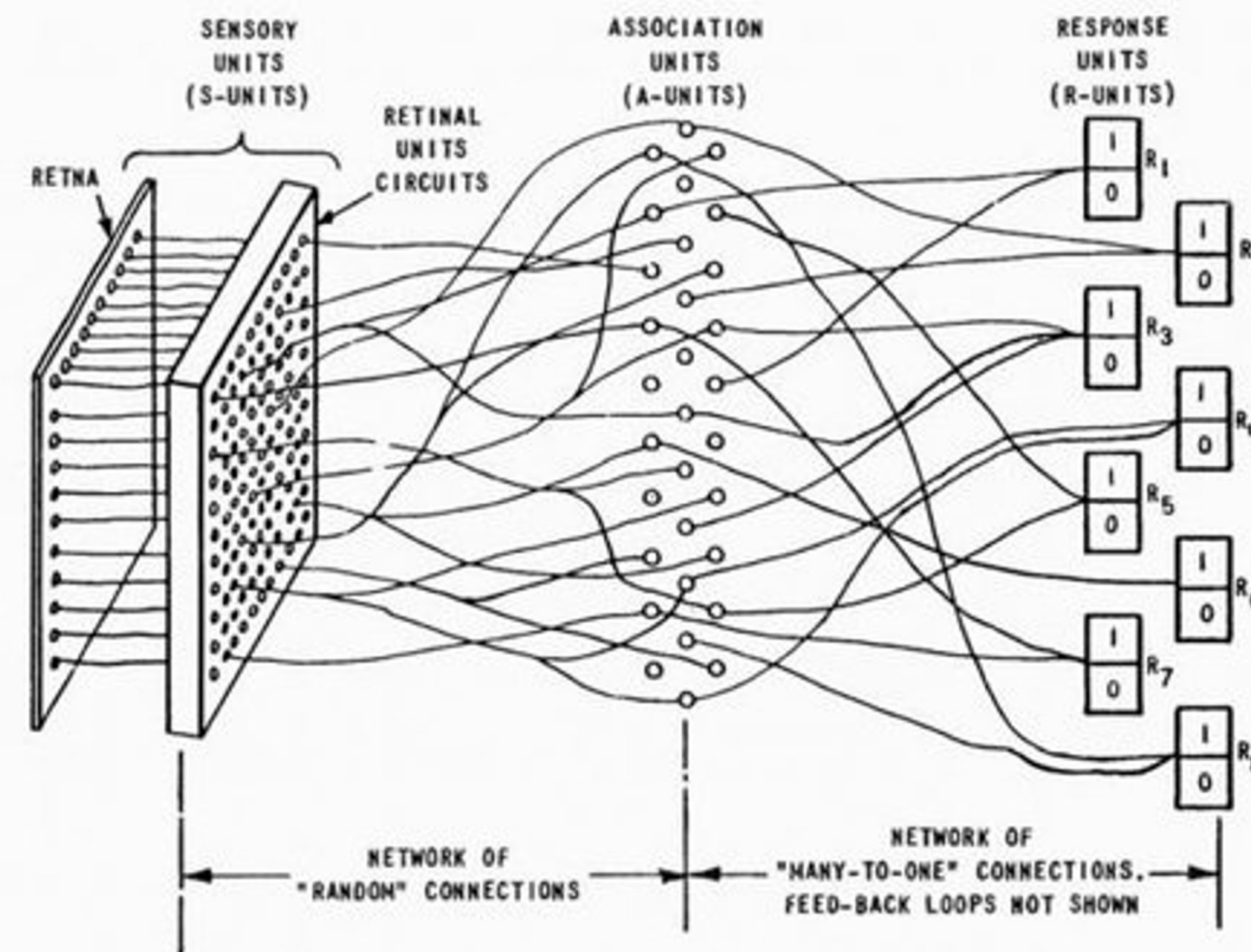
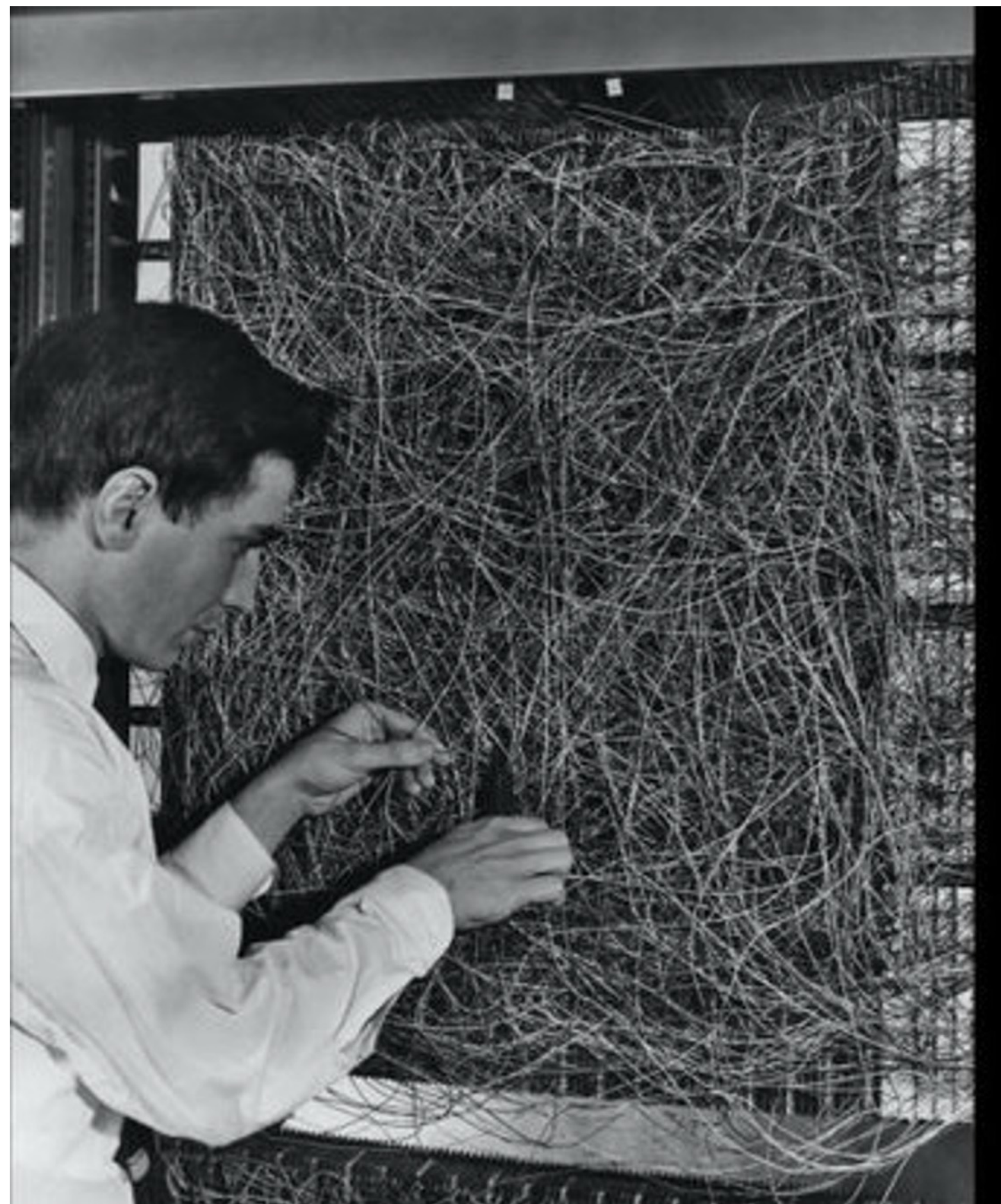


Figure 1 ORGANIZATION OF THE MARK I PERCEPTRON

Why are we here?



- **What's the goal?**
 - Solve complex problems in the real world
- **What are complex problems?**
 - Natural language, Images, Robotics, Games, DNA, Molecules, ...
- **Existence proof for deep learning as a solution:** The human brain?

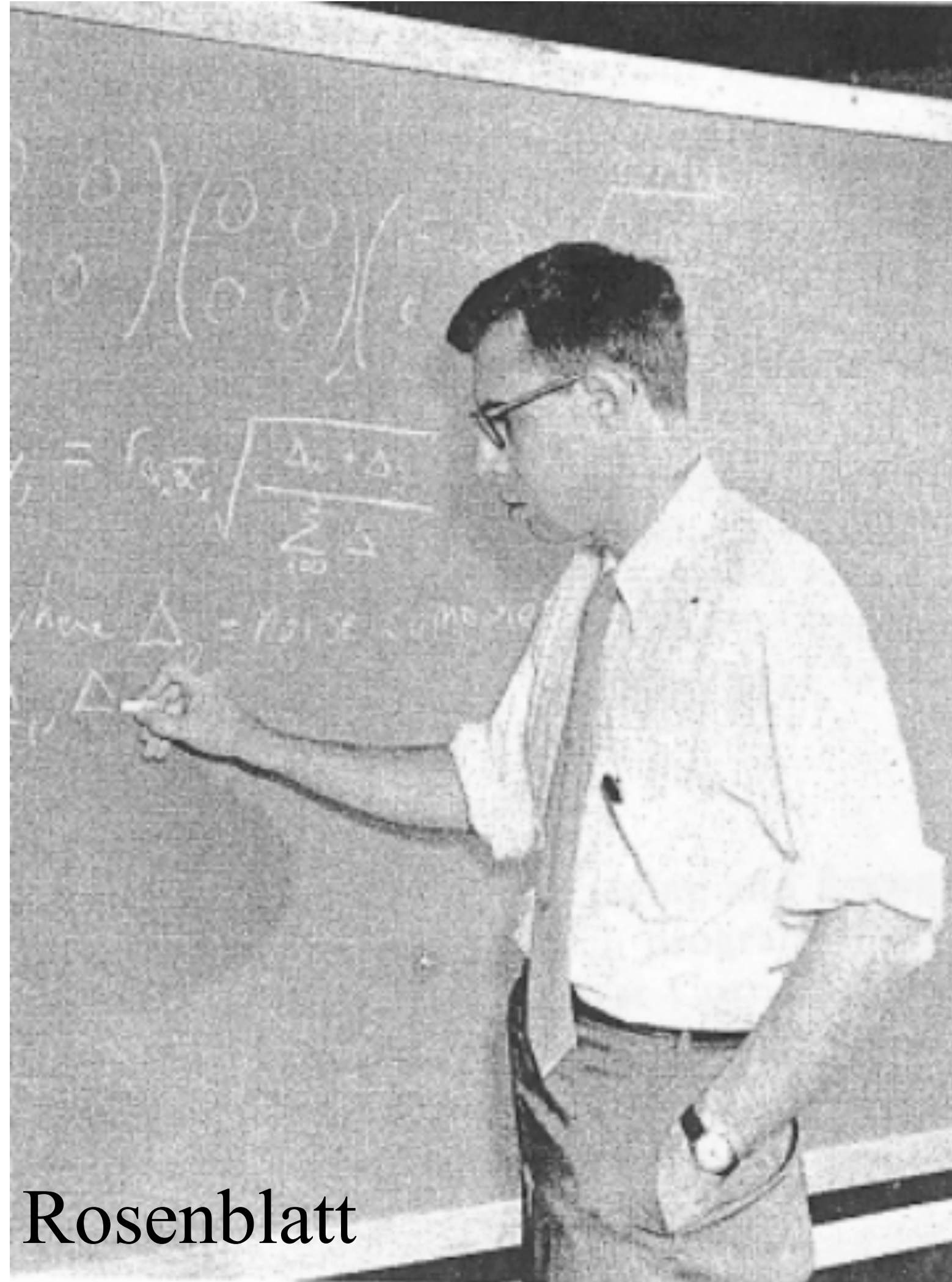
1. Introduction to Deep Learning

- A Brief History of (Artificial) Neural Networks
- A Quick Look at Deep Neural Networks

A brief history of Neural Networks

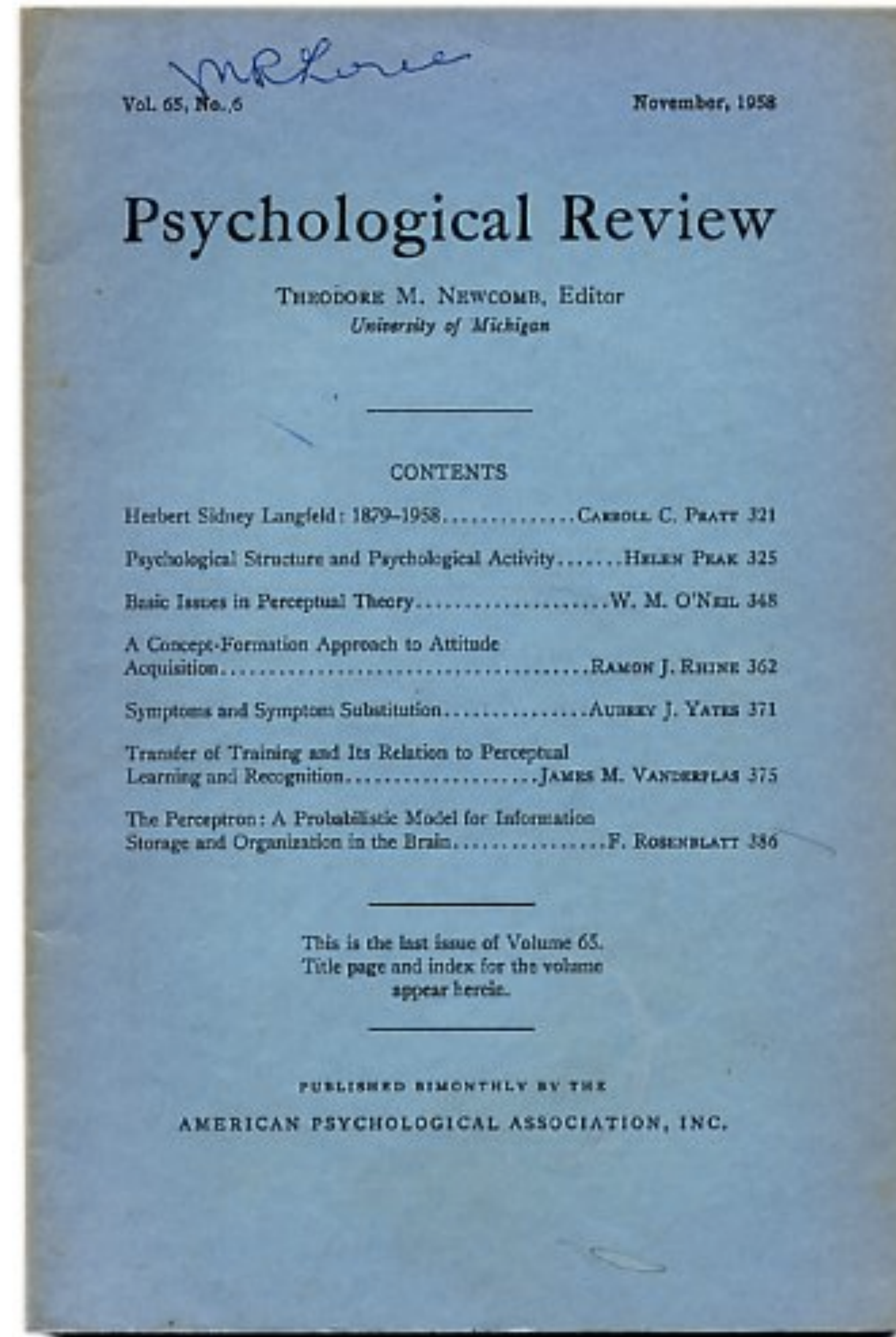


Perceptrons, 1958



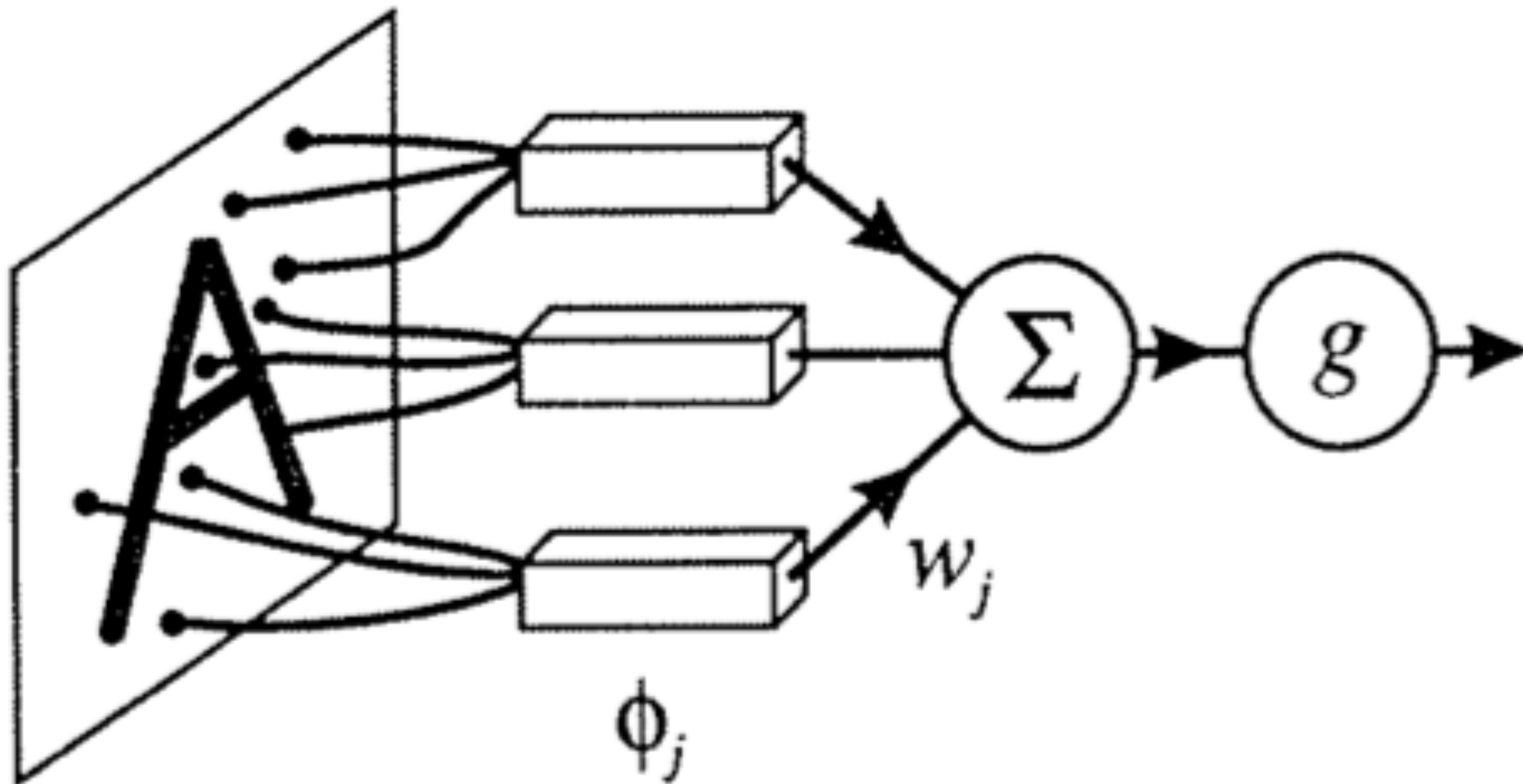
Rosenblatt

http://www.ecse.rpi.edu/homepages/nagy/PDF_chrono/2011_Nagy_Pace_FR.pdf. Photo by George Nagy



<http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.335.3398&rep=rep1&type=pdf>

Perceptrons, 1958



Perceptrons, 1958

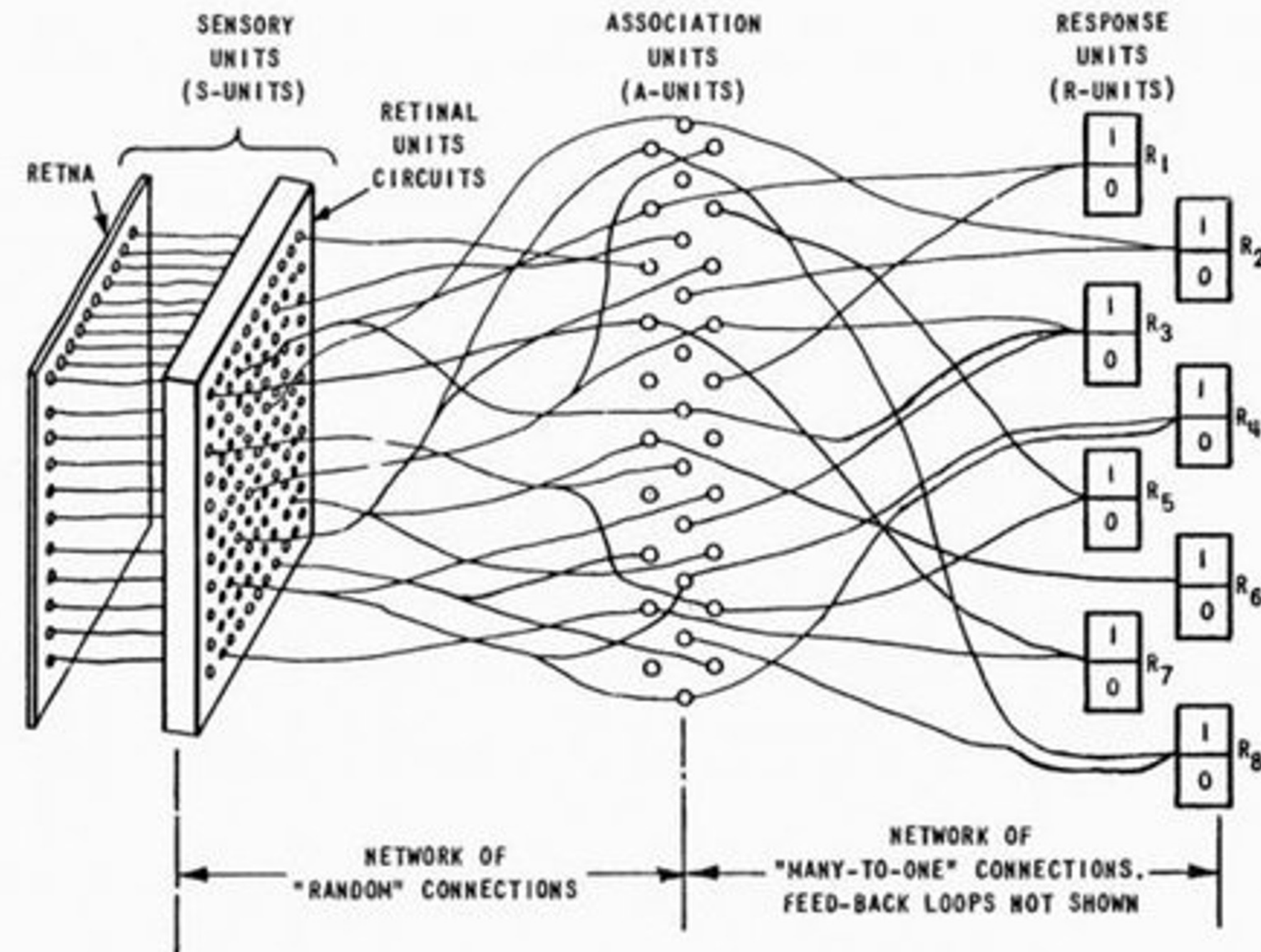
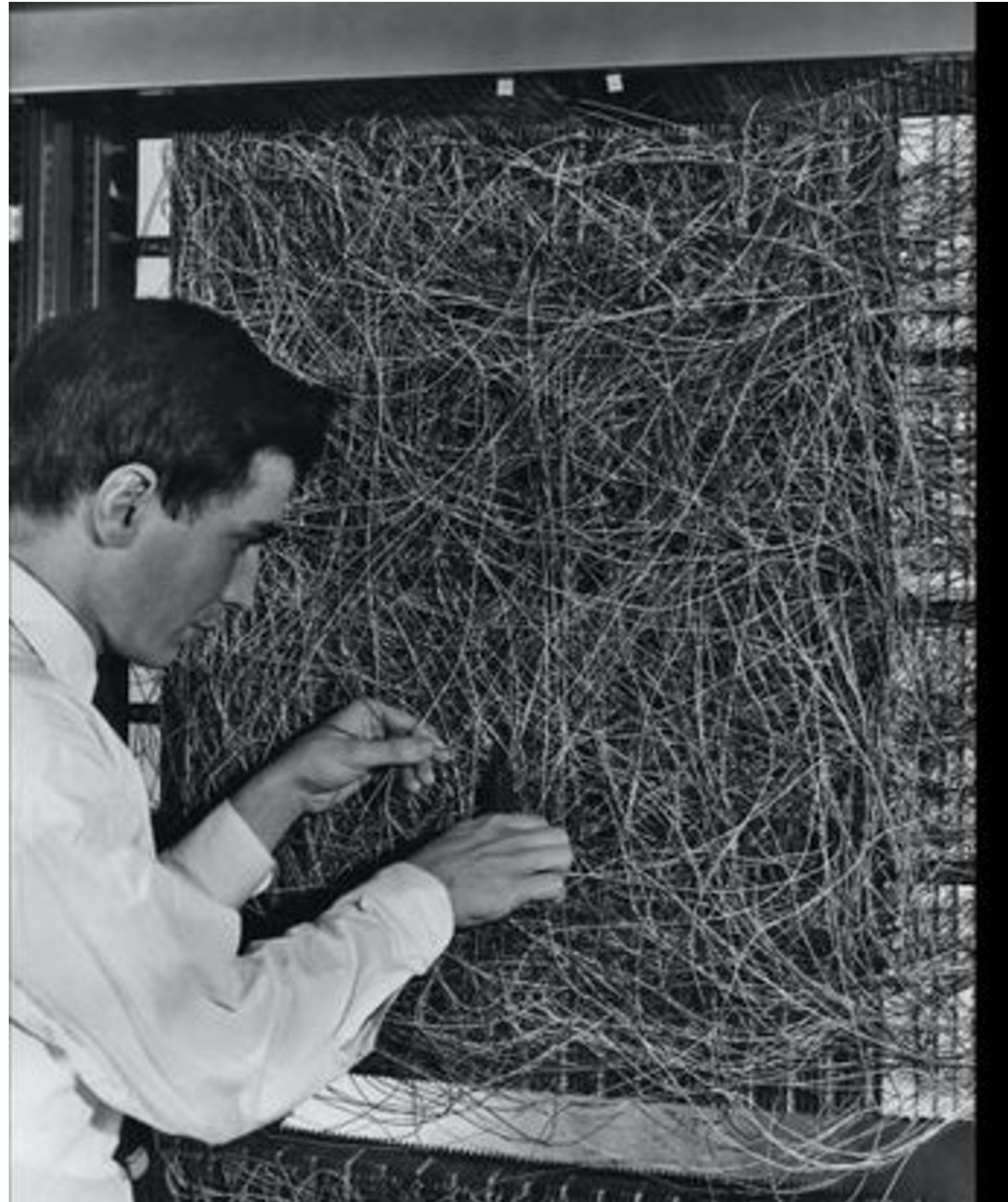
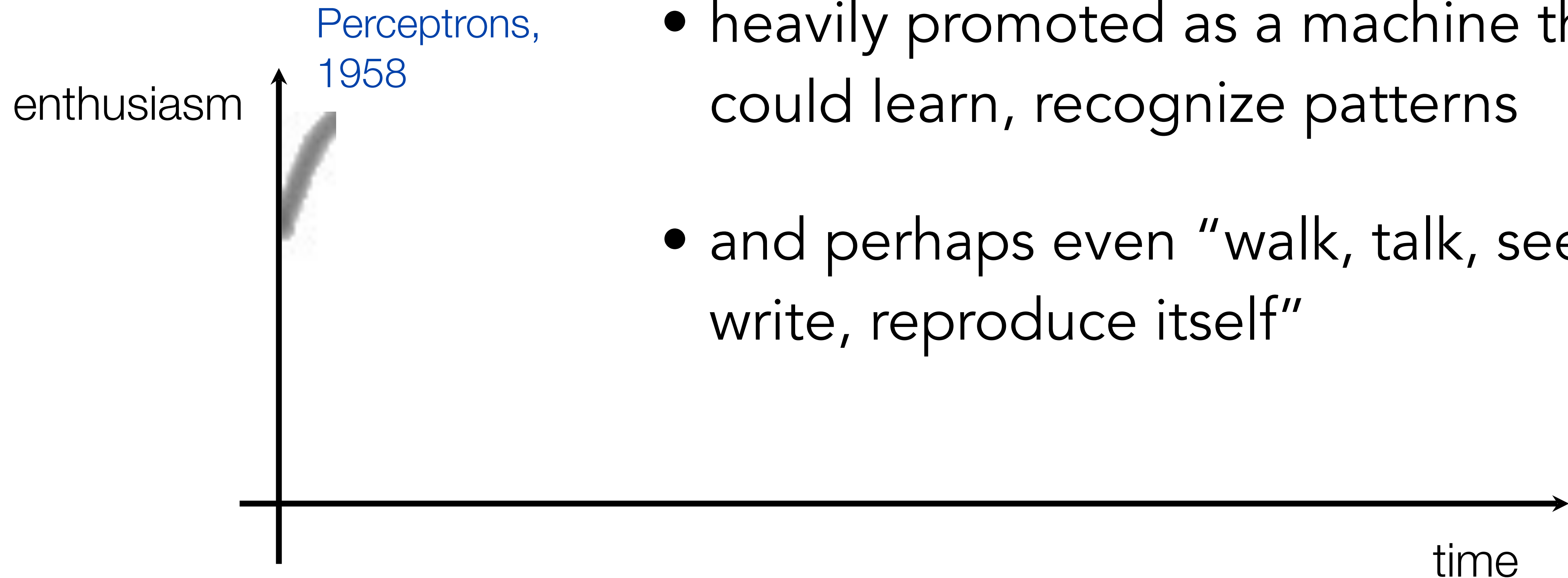


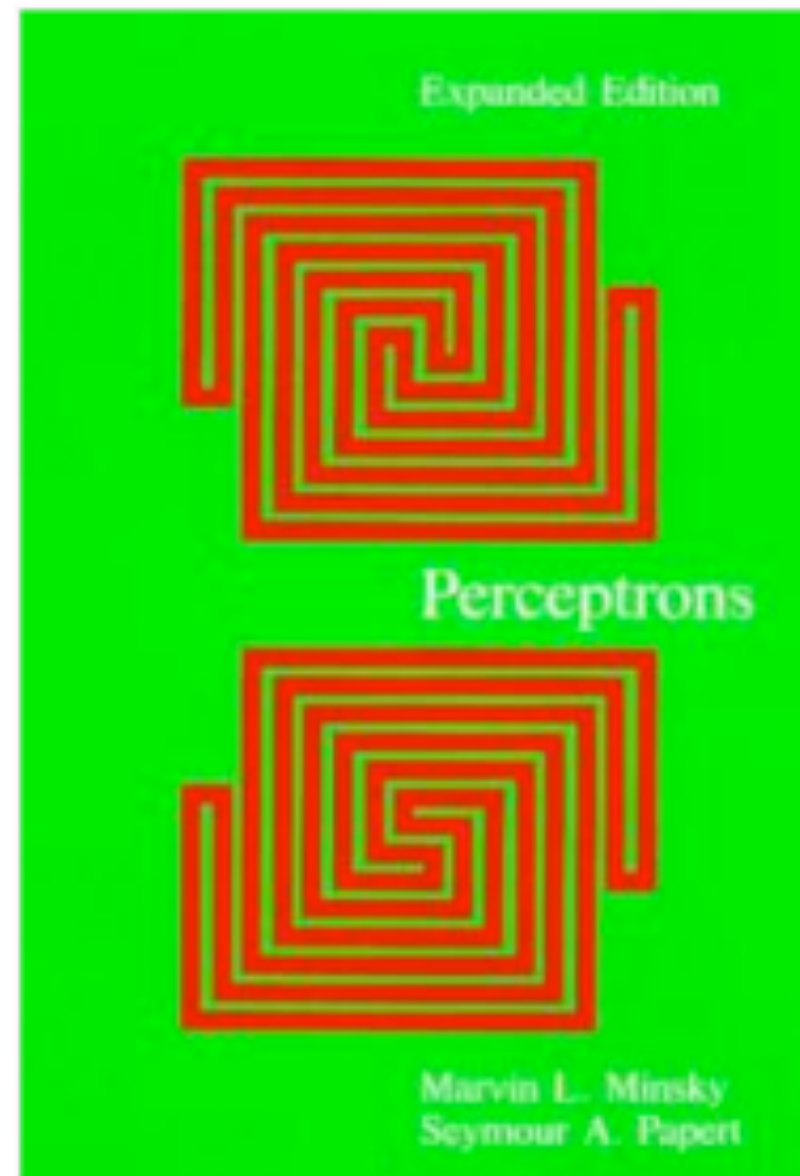
Figure 1 ORGANIZATION OF THE MARK I PERCEPTRON

Media hype was extreme

- heavily promoted as a machine that could learn, recognize patterns
- and perhaps even “walk, talk, see, write, reproduce itself”



Minsky and Papert, Perceptrons, 1969



FOR BUYING OPTIONS, START HERE

Select Shipping Destination

Paperback | \$35.00 Short | £24.95 |
ISBN: 9780262631112 | 308 pp. | 6 x
8.9 in | December 1987

Perceptrons, expanded edition

An Introduction to Computational Geometry

By Marvin Minsky and Seymour A. Papert

Overview

Perceptrons - the first systematic study of parallelism in computation - has remained a classical work on threshold automata networks for nearly two decades. It marked a historical turn in artificial intelligence, and it is required reading for anyone who wants to understand the connectionist counterrevolution that is going on today.

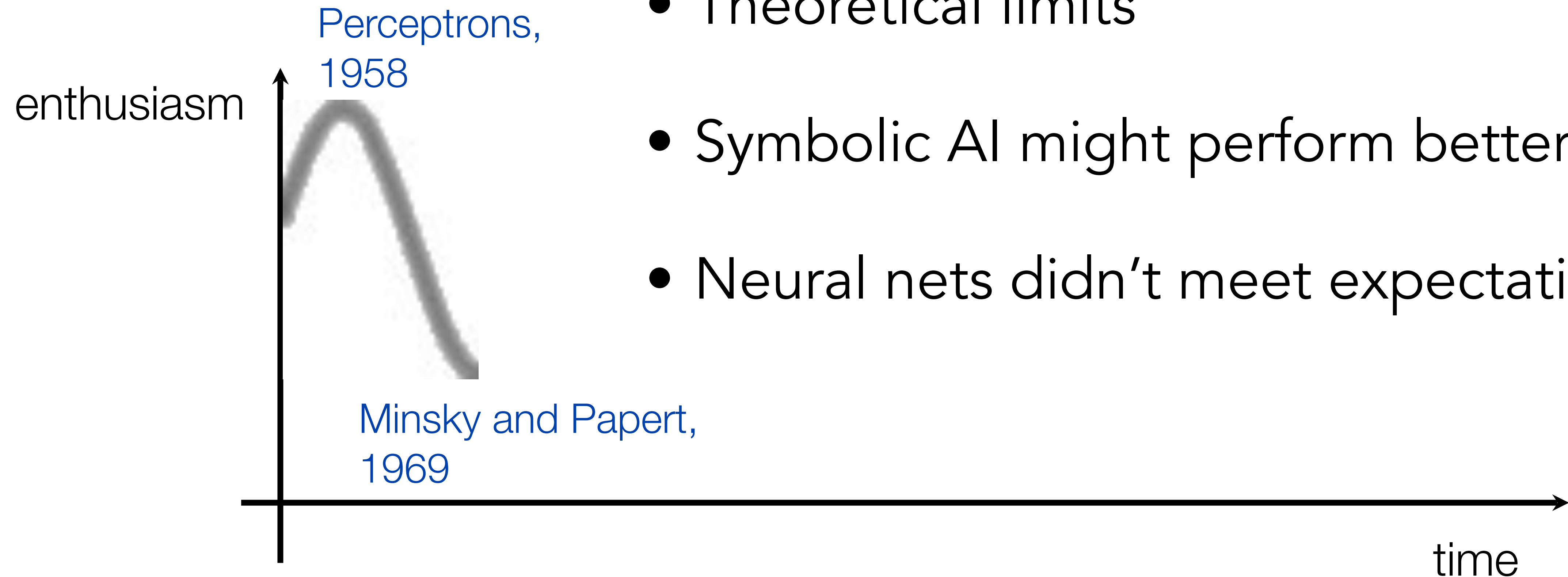
Artificial-intelligence research, which for a time concentrated on the programming of von Neumann computers, is swinging back to the idea that intelligence might emerge from the activity of networks of neuronlike entities. Minsky and Papert's book was the first example of a mathematical analysis carried far enough to show the exact limitations of a class of computing machines that could seriously be considered as models of the brain. Now the new developments in mathematical tools, the recent interest of physicists in the theory of disordered matter, the new insights into and psychological models of how the brain works, and the evolution of fast computers that can simulate networks of automata have given *Perceptrons* new importance.

Witnessing the swing of the intellectual pendulum, Minsky and Papert have added a new chapter in which they discuss the current state of parallel computers, review developments since the appearance of the 1972 edition, and identify new research directions related to connectionism. They note a central theoretical challenge facing connectionism: the challenge to reach a deeper understanding of how "objects" or "agents" with individuality can emerge in a network. Progress in this area would link connectionism with what the authors have called "society theories of mind."

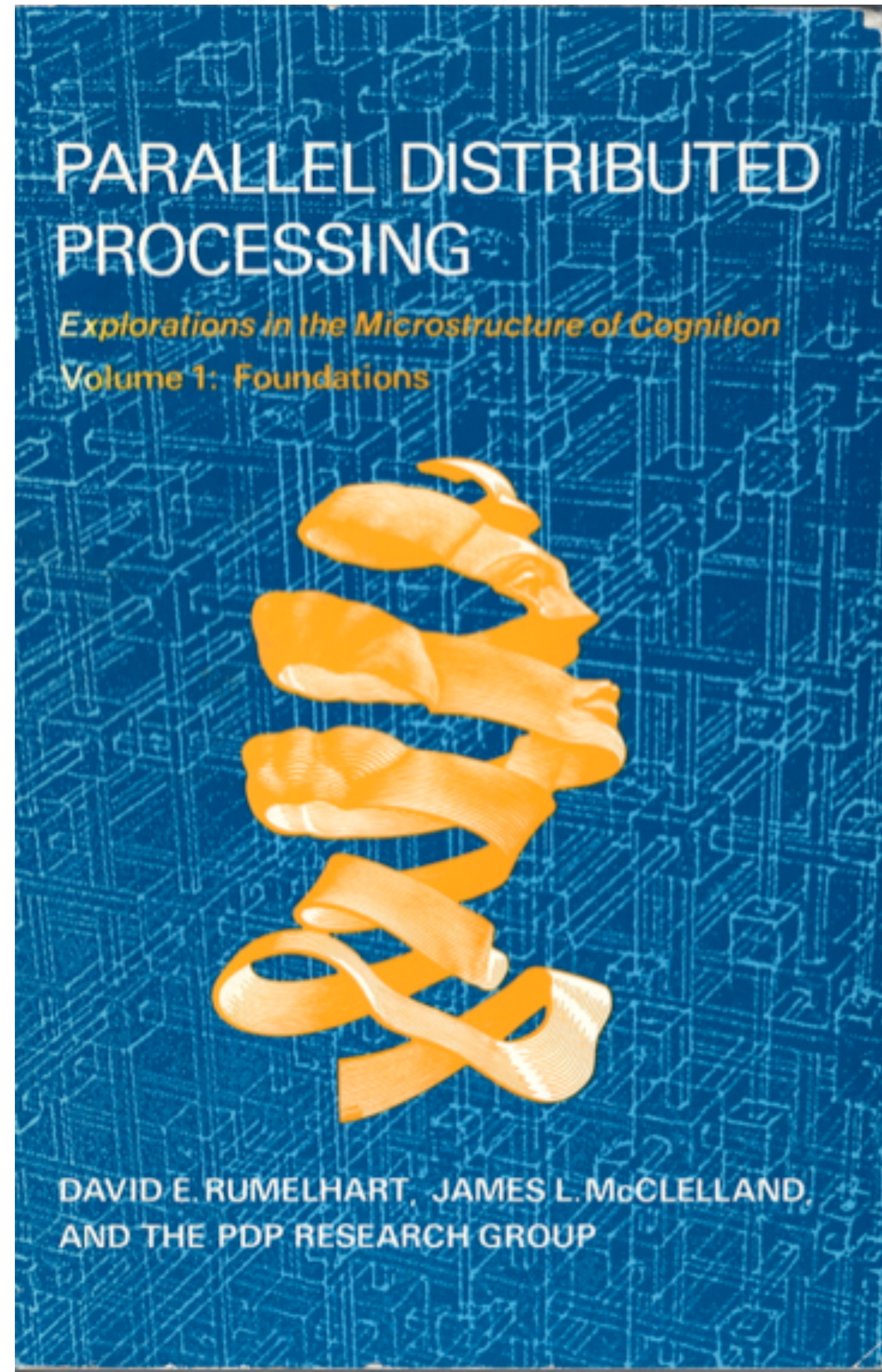
The XOR limitation

Pessimism about neural nets

- Theoretical limits
- Symbolic AI might perform better
- Neural nets didn't meet expectations



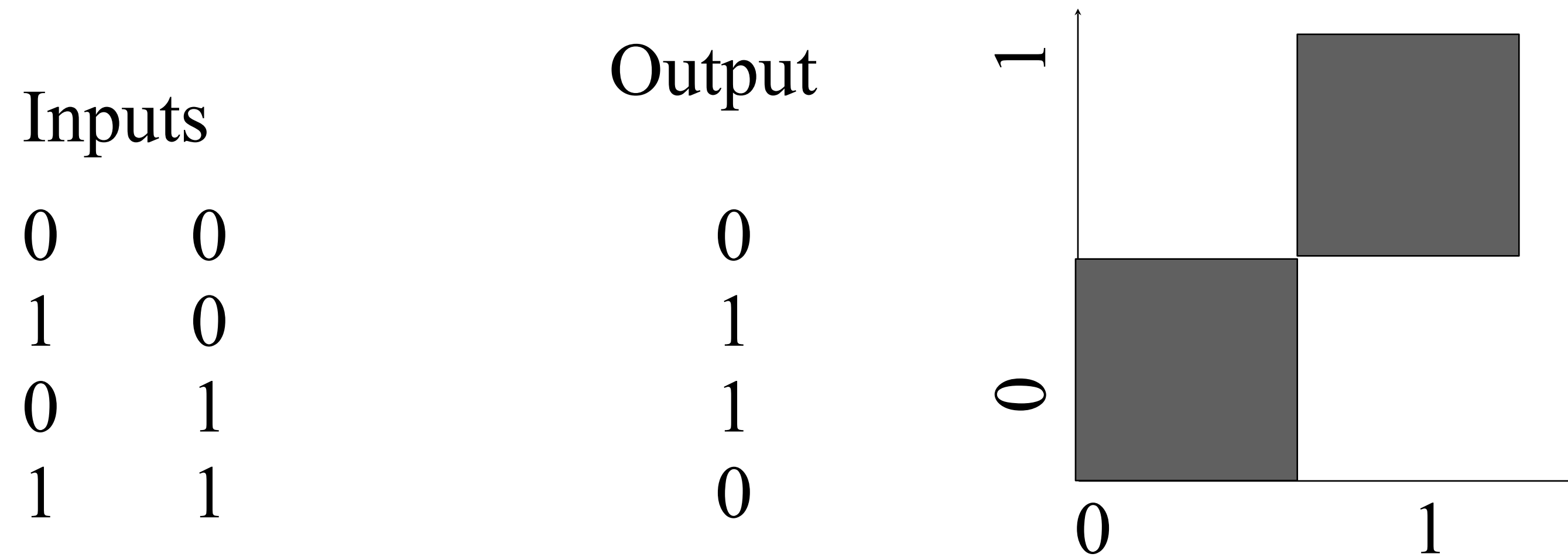
Parallel Distributed Processing (PDP), 1986



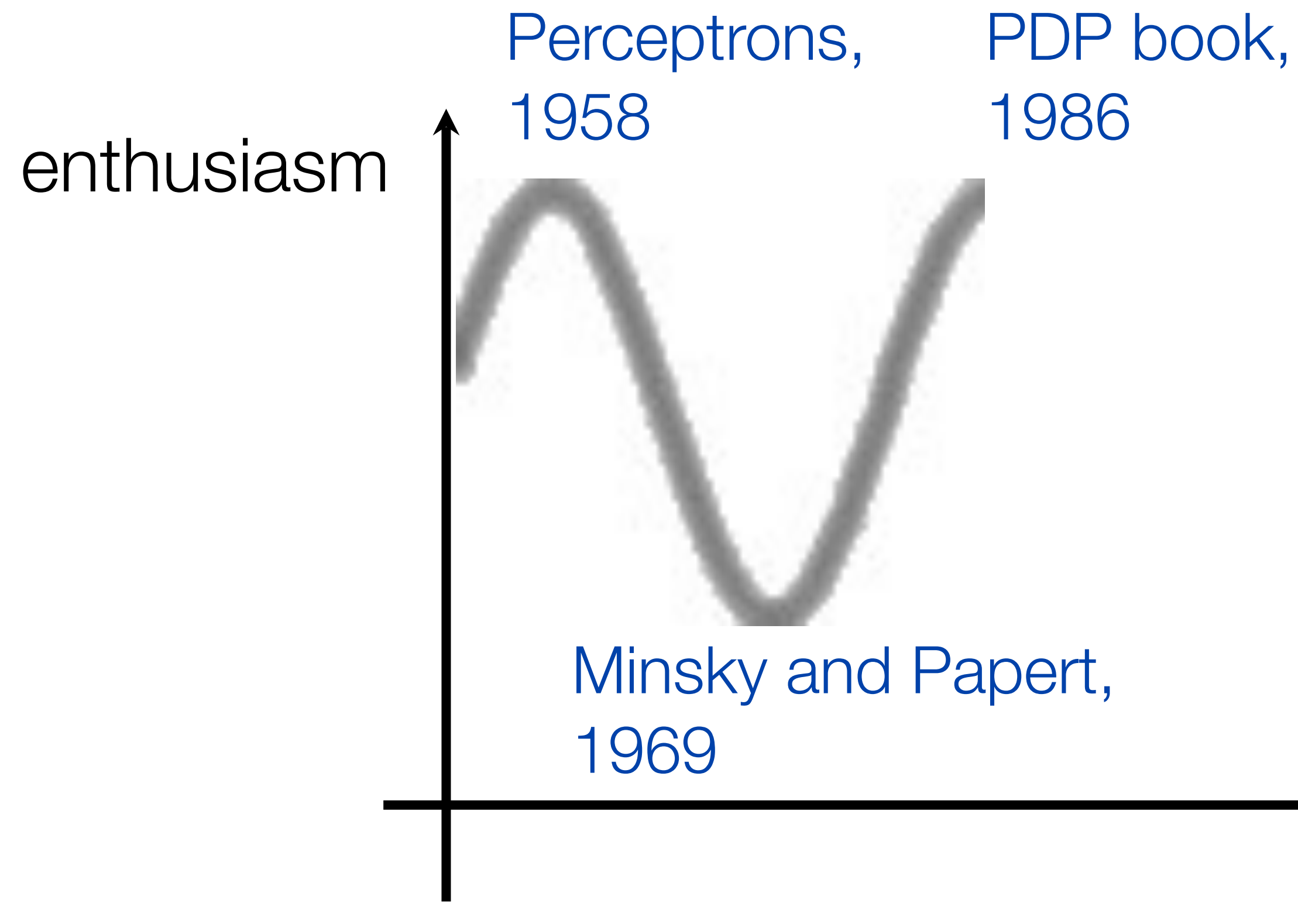
“Parallel” + “Distributed”

- The primary emphasis was not on “depth”.
- Many simple and interconnected units (neurons) act in parallel
- Representations are distributed across units

XOR problem



PDP authors pointed to the **backpropagation** algorithm as a breakthrough, allowing **multi-layer neural networks** to be trained. Among the functions that a multi-layer network can represent but a single-layer network cannot: the XOR function.



A paradigm reversal

- Backprop got popularized
- Multilayer nets shown to be useful
- Evidence of learning complex functions

LeCun ConvNets, 1989, 1998

PROC. OF THE IEEE, NOVEMBER 1998

7

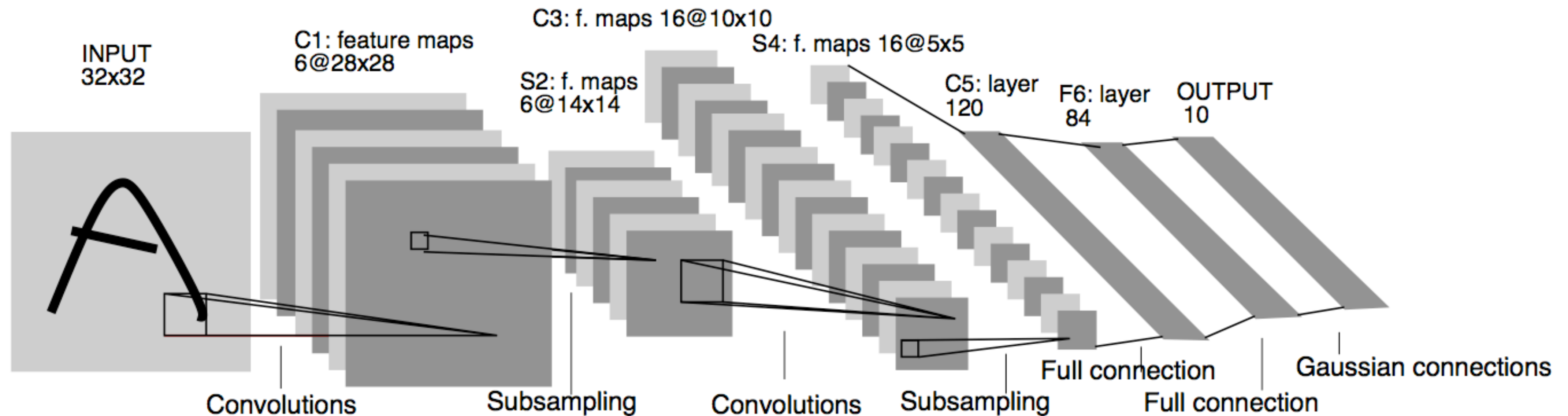
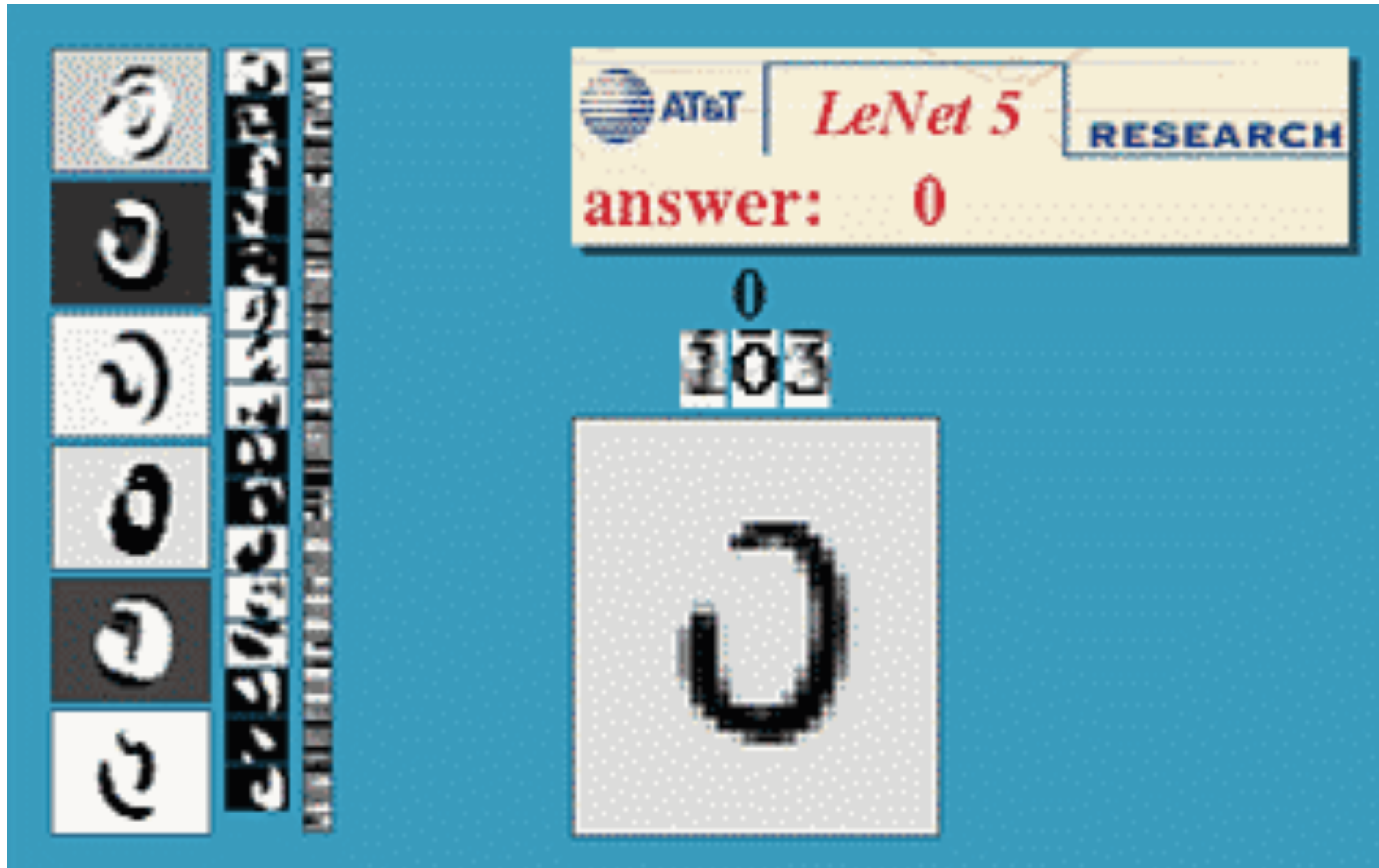


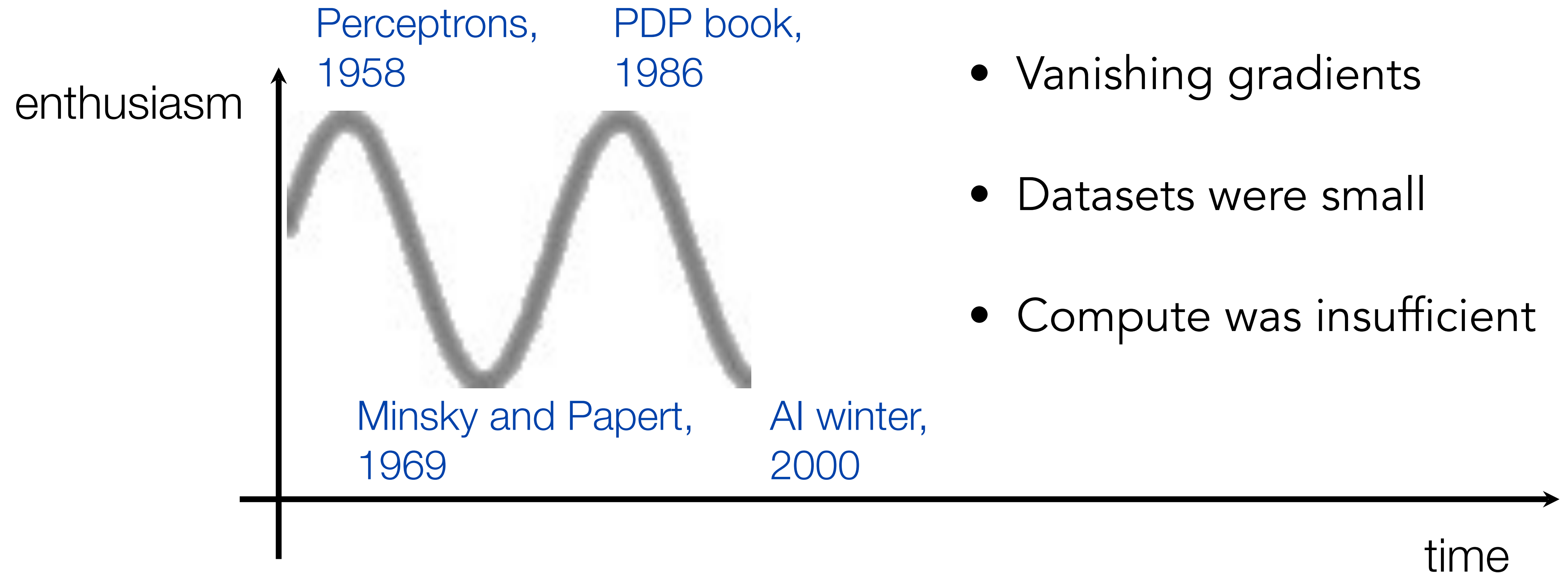
Fig. 2. Architecture of LeNet-5, a Convolutional Neural Network, here for digits recognition. Each plane is a feature map, i.e. a set of units whose weights are constrained to be identical.

Demos:

<http://yann.lecun.com/exdb/lenet/index.html>

LeCun ConvNets, 1989, 1998





Hit the next wall:

- Training was unstable
- Vanishing gradients
- Datasets were small
- Compute was insufficient

Support Vector Machines (SVMs) and Decision Trees

- more popular than Neural Nets back then
- easier and faster to train
- worked better with limited data

Machine Learning, 20, 273–297 (1995)
© 1995 Kluwer Academic Publishers, Boston. Manufactured in The Netherlands.

Support-Vector Networks

CORINNA CORTES
VLADIMIR VAPNIK
AT&T Bell Labs., Holmdel, NJ 07733, USA

corinna@neural.att.com
vlad@neural.att.com

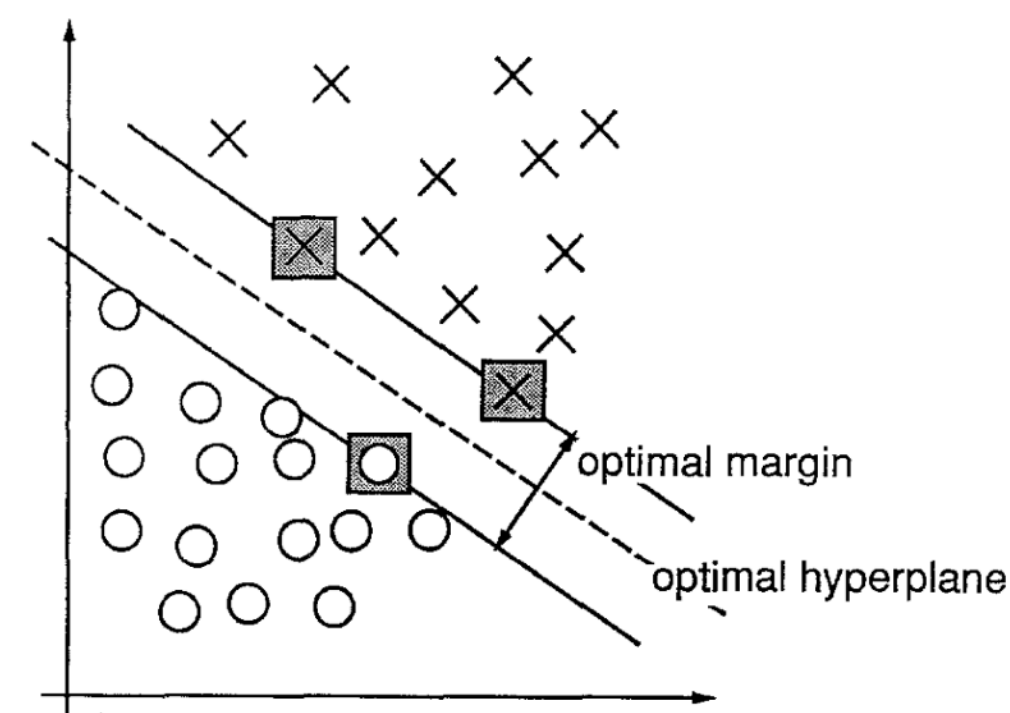
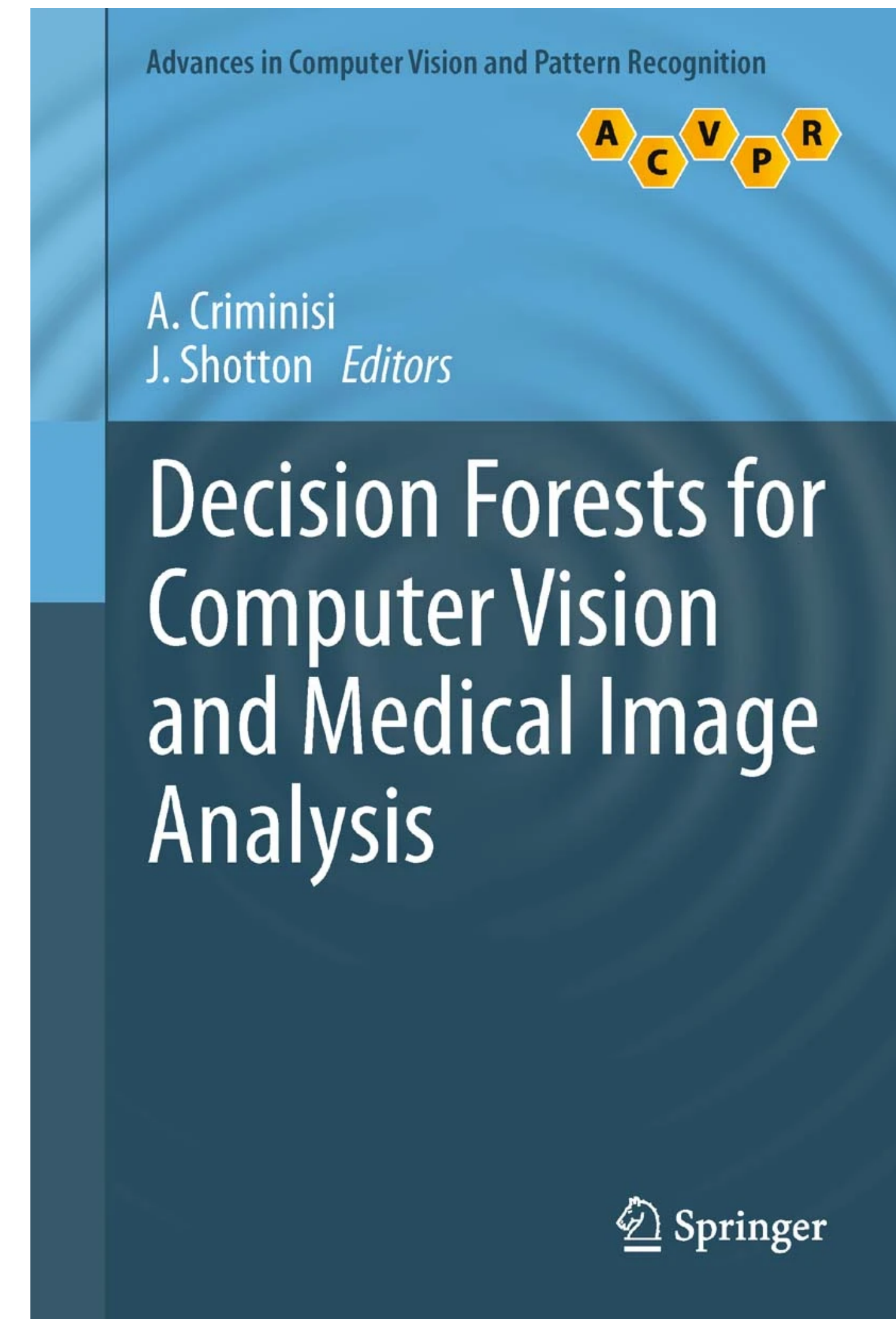
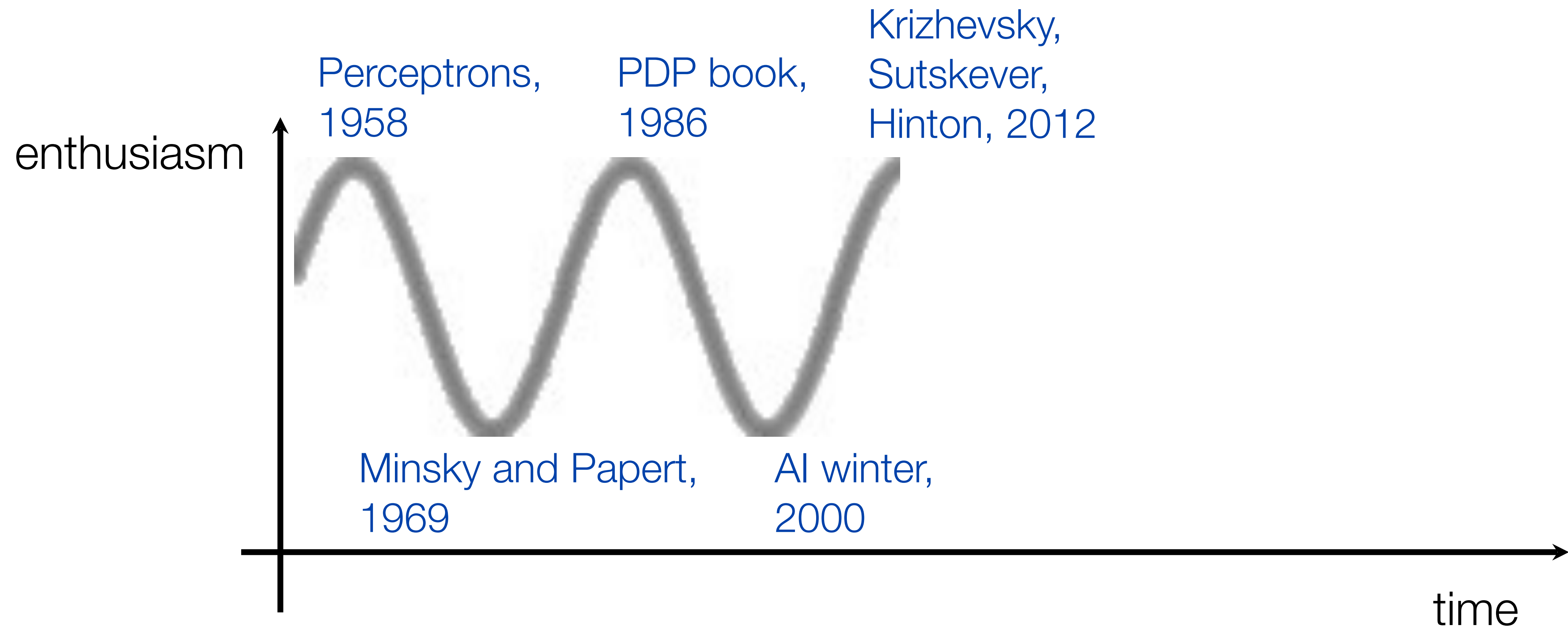


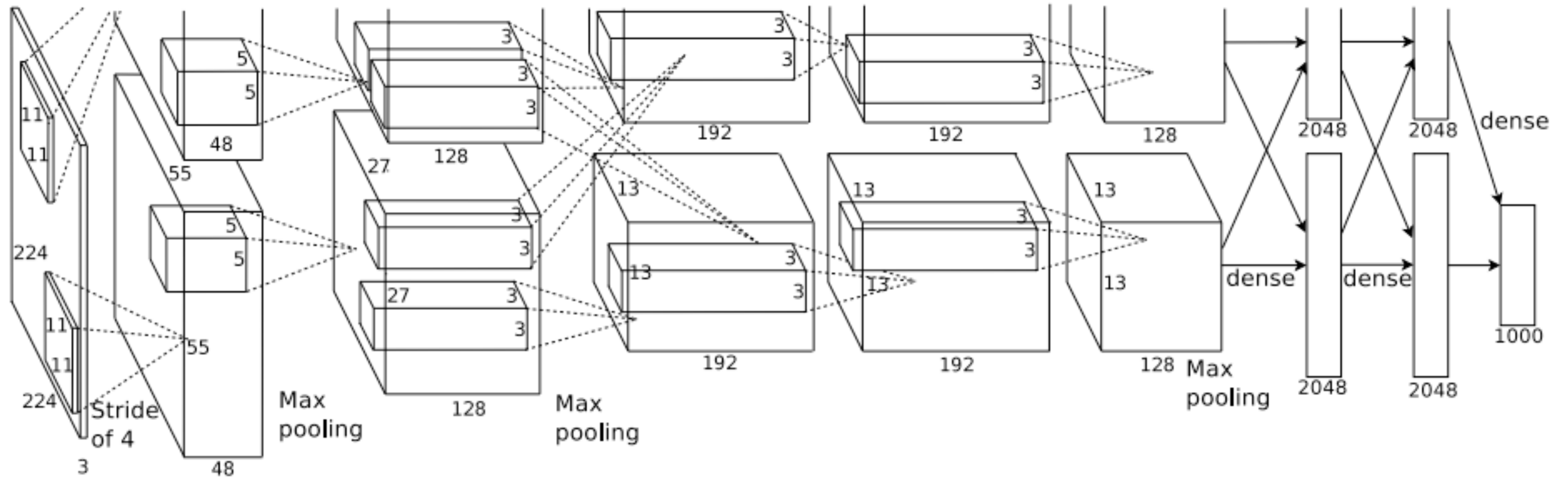
Figure 2. An example of a separable problem in a 2 dimensional space. The support vectors, marked with grey squares, define the margin of largest separation between the two classes.





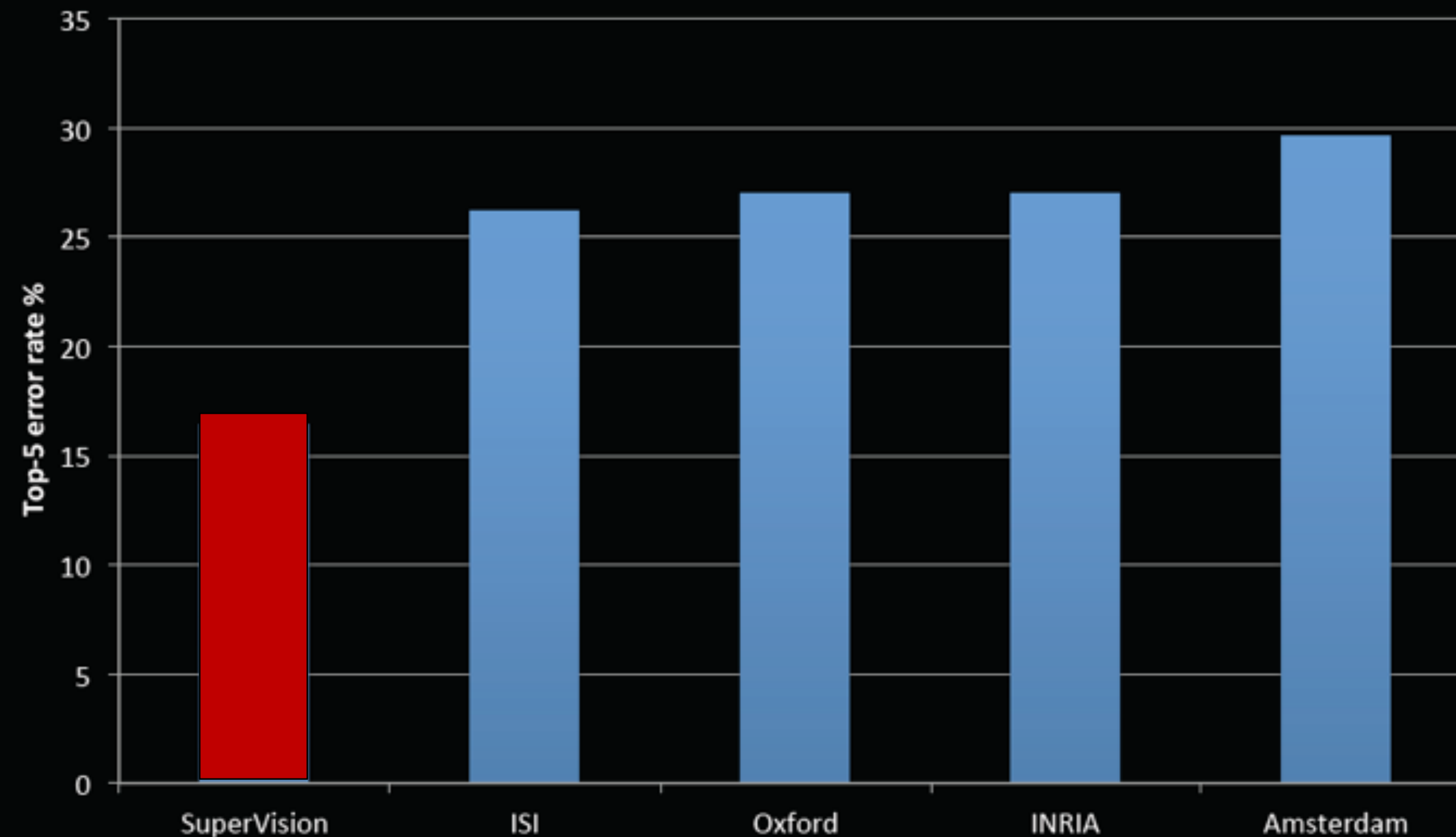
Krizhevsky, Sutskever, and Hinton, NeurIPS 2012

“AlexNet”

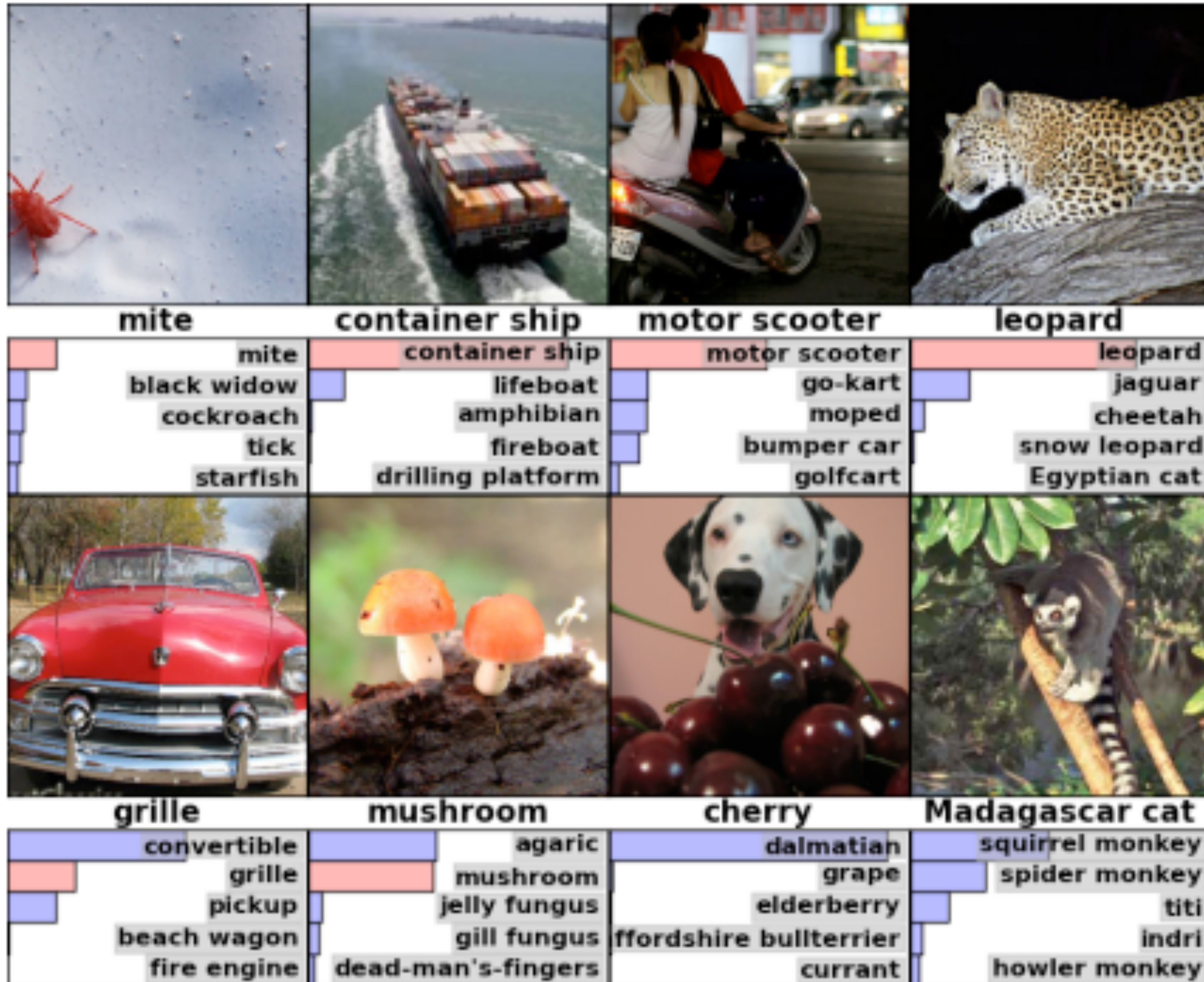


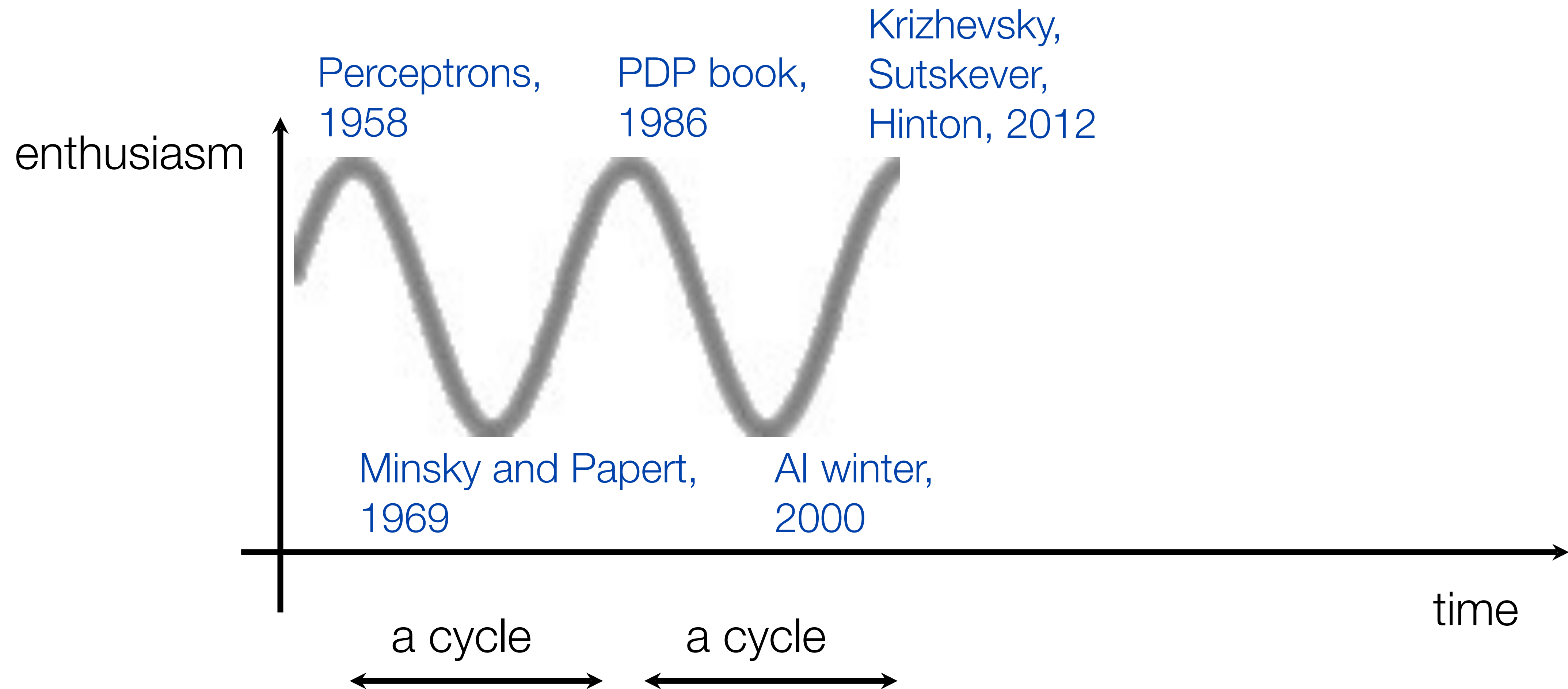
ImageNet Classification 2012

- Krizhevsky et al. -- 16.4% error (top-5)
- Next best (non-convnet) – 26.2% error

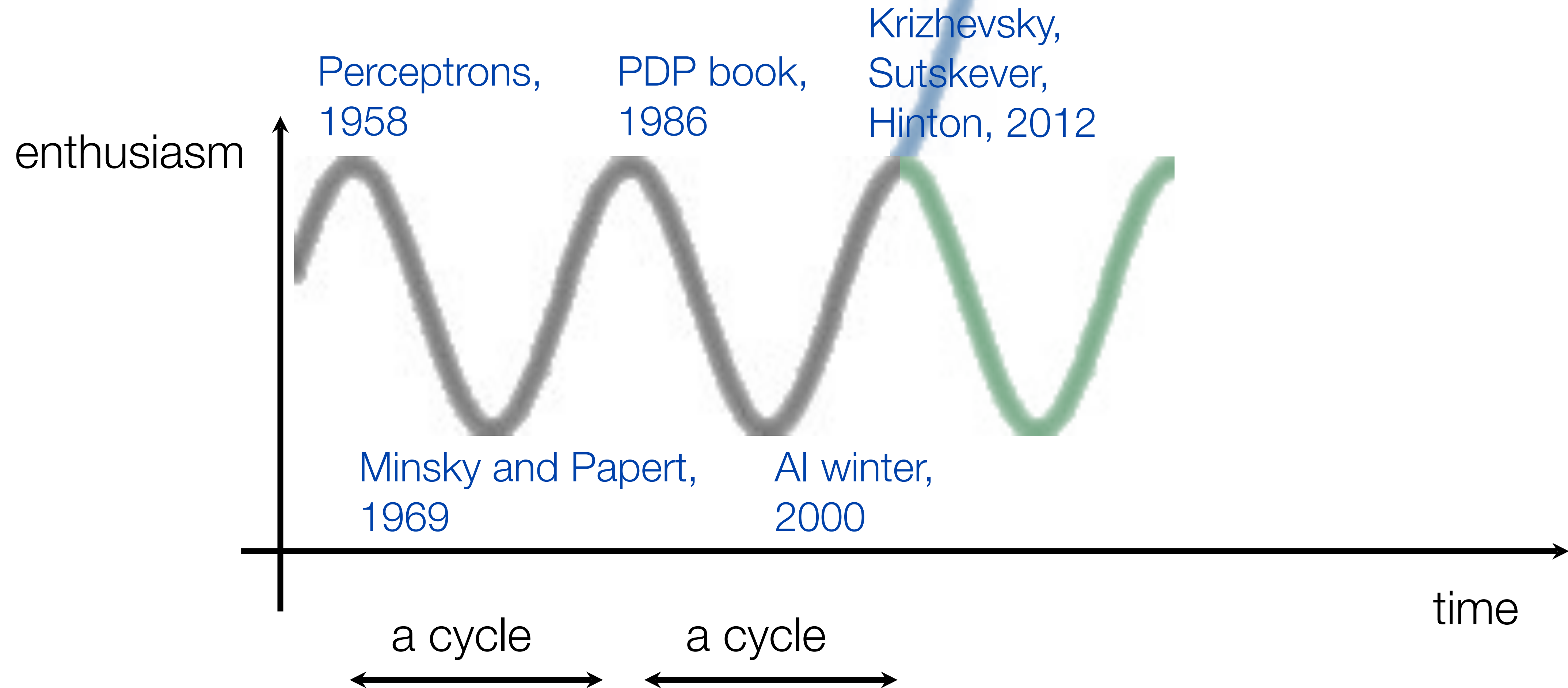


Krizhevsky, Sutskever, and Hinton, NeurIPS 2012



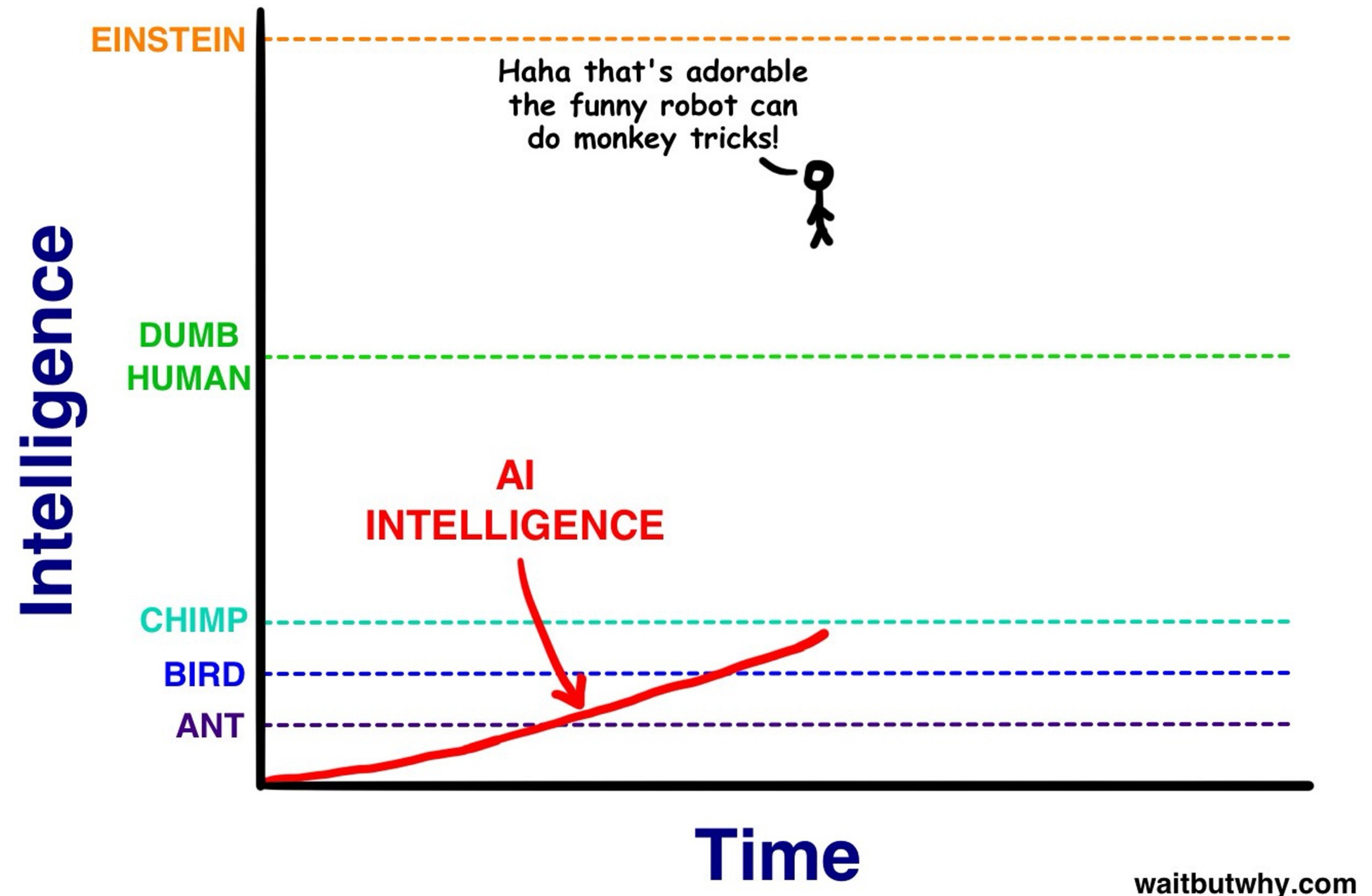


What comes next?



A perspective
[Tim Urban, 2015
“Wait But Why”]

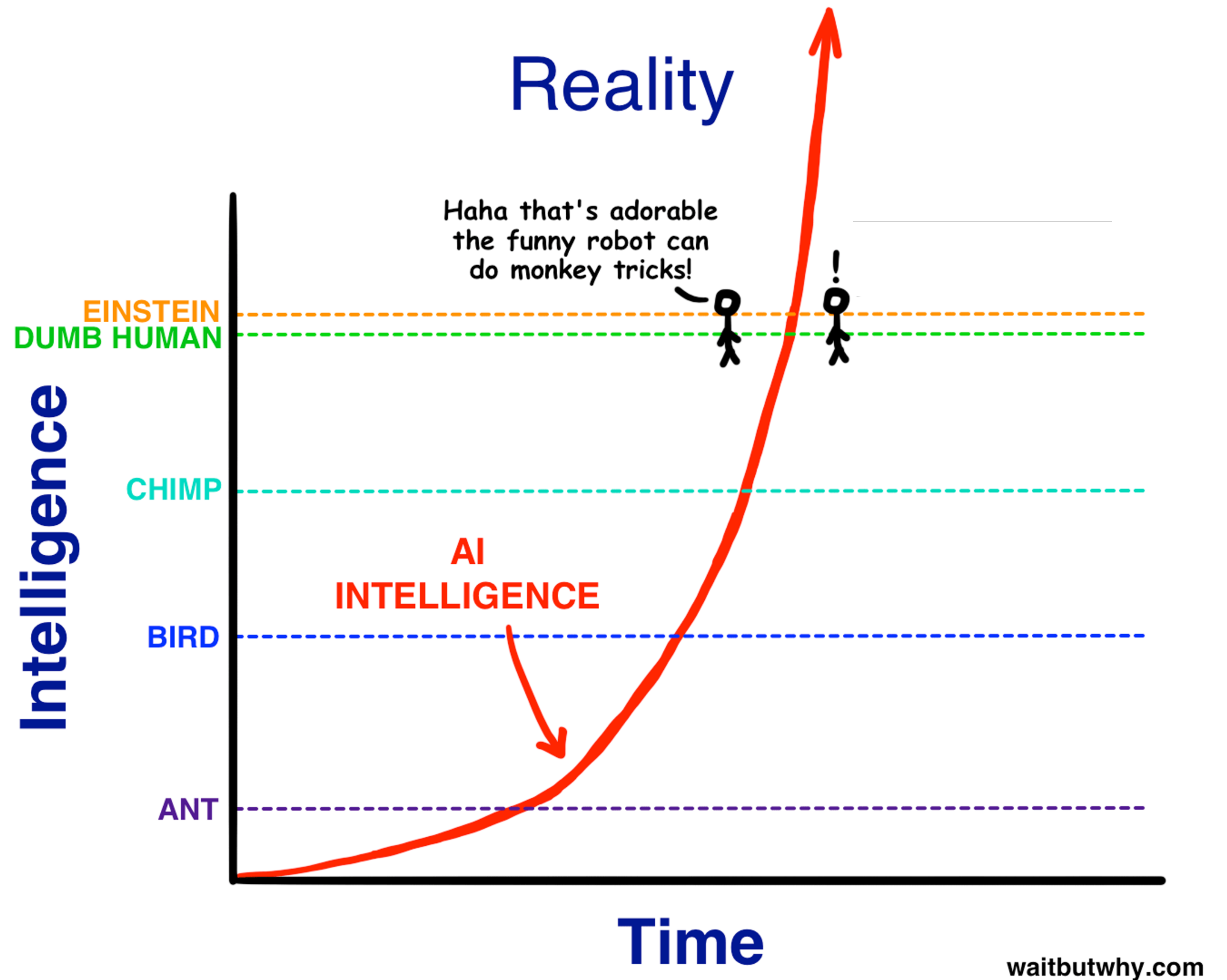
Our Distorted View of Intelligence



waitbutwhy.com

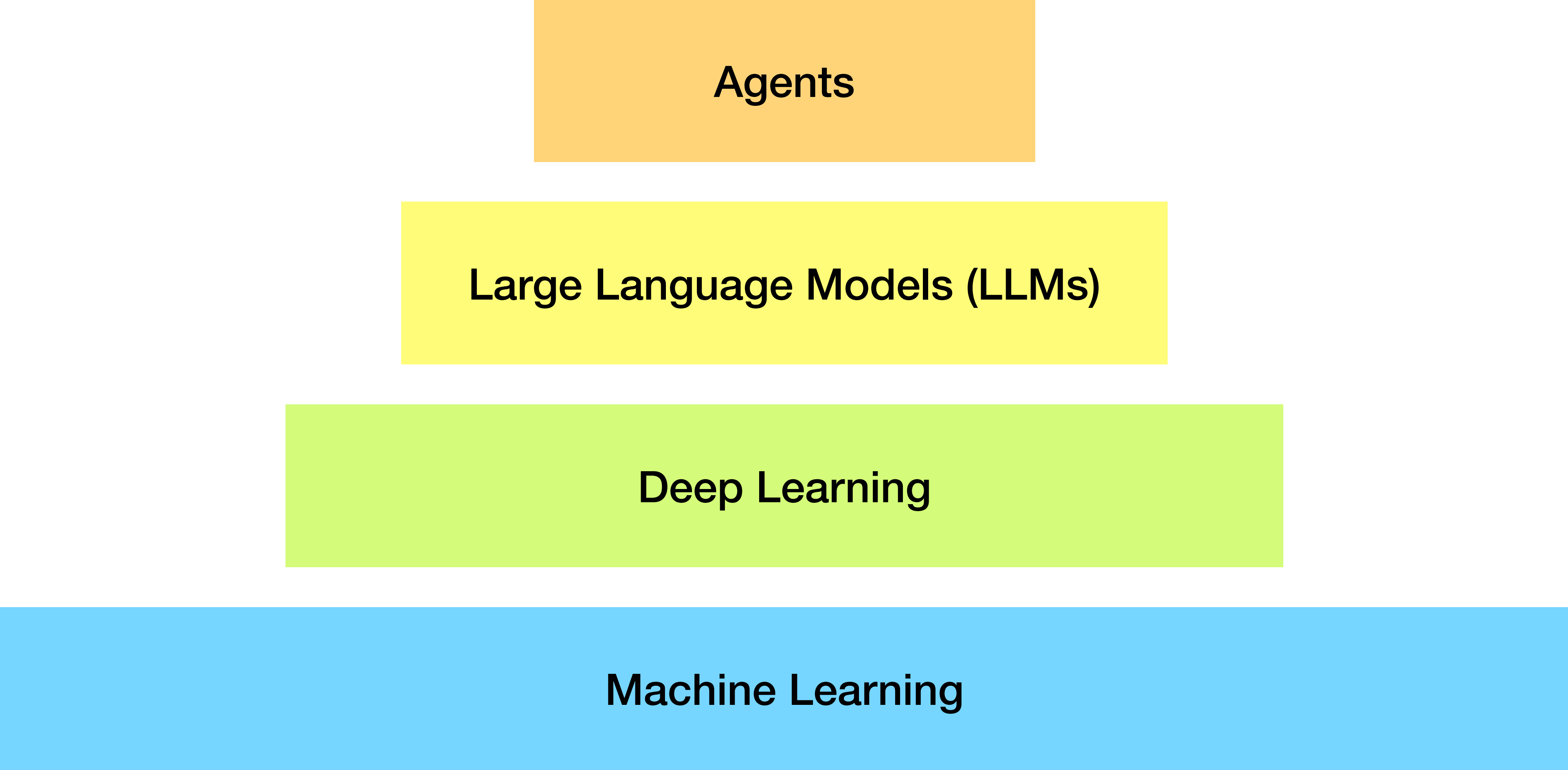
<https://maartensteinbuch.com/2015/02/02/the-ai-revolution-the-road-to-superintelligence/>

A perspective
[Tim Urban, 2015
“Wait But Why”]



What is deep learning today?

Agents

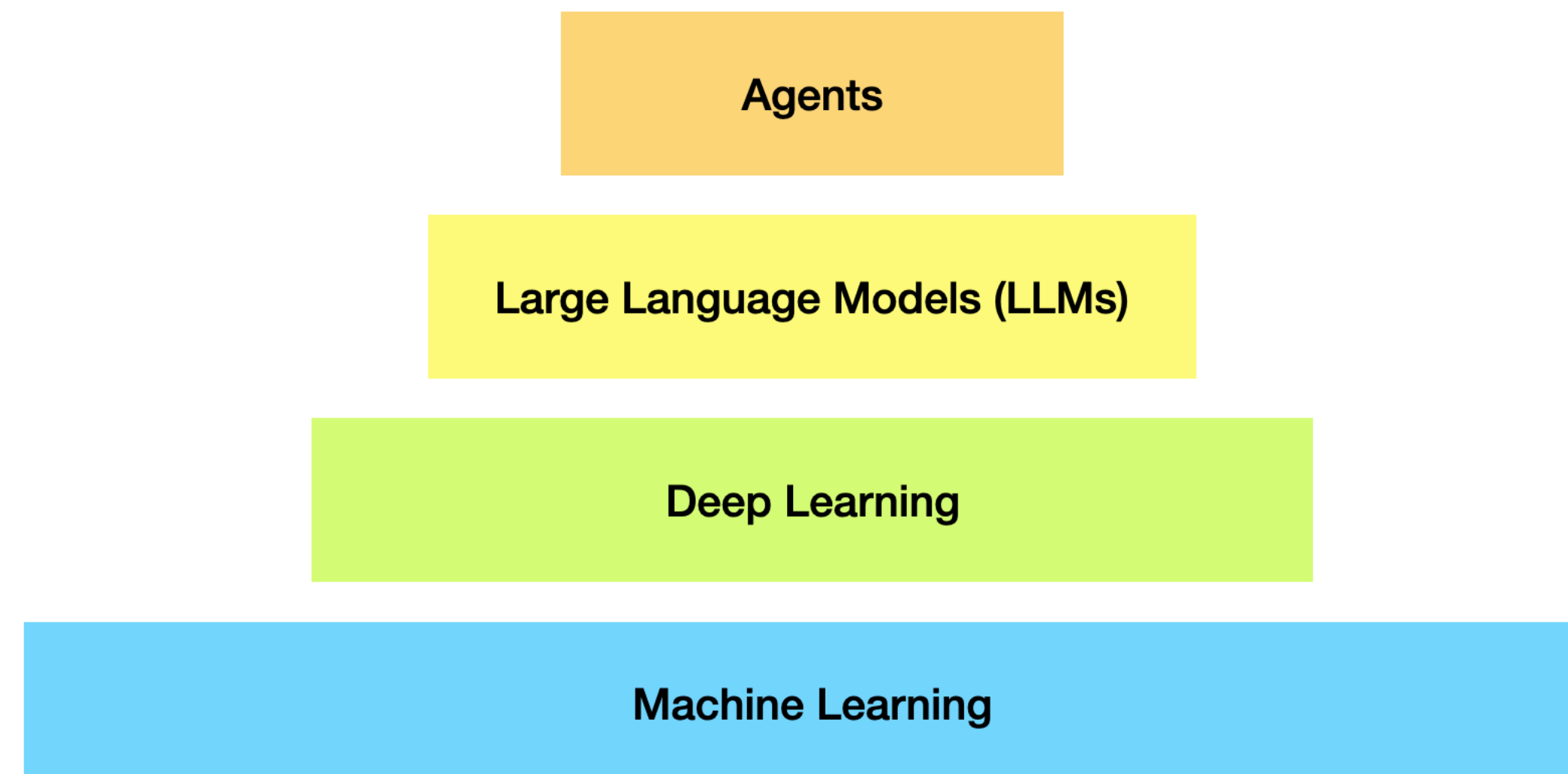


Large Language Models (LLMs)

Deep Learning

Machine Learning

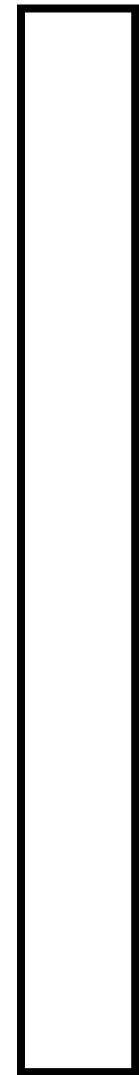
- ML → DL → LLM → Agent:
Each develops into a field of study in its own right.
- Analogous to physics → chemistry → biology:
Each field builds on the foundations below it, while developing its own research questions and methods.



A Quick Look at Deep Neural Networks

Computation in a neural net

Input
representation

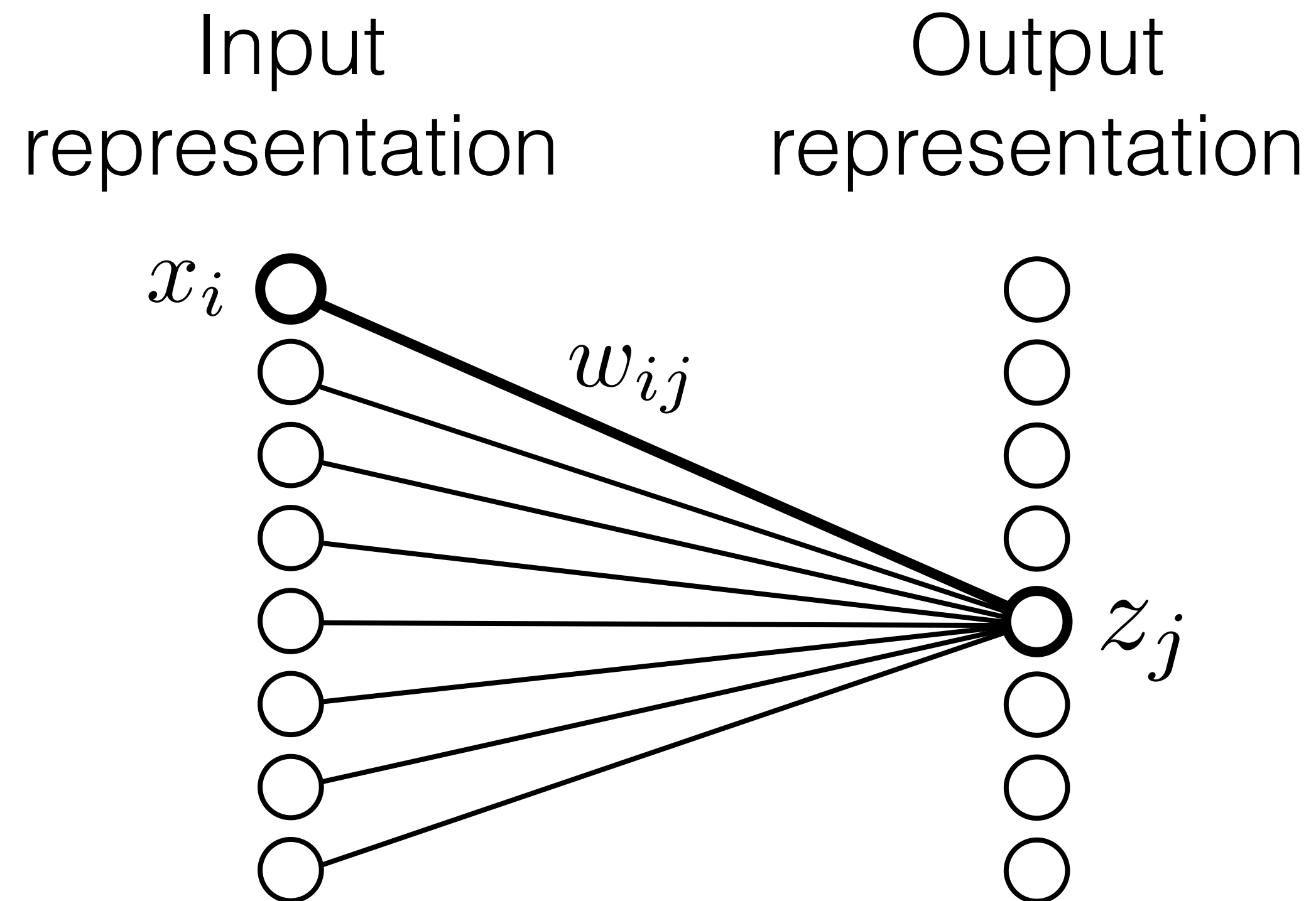


Output
representation



Computation in a neural net

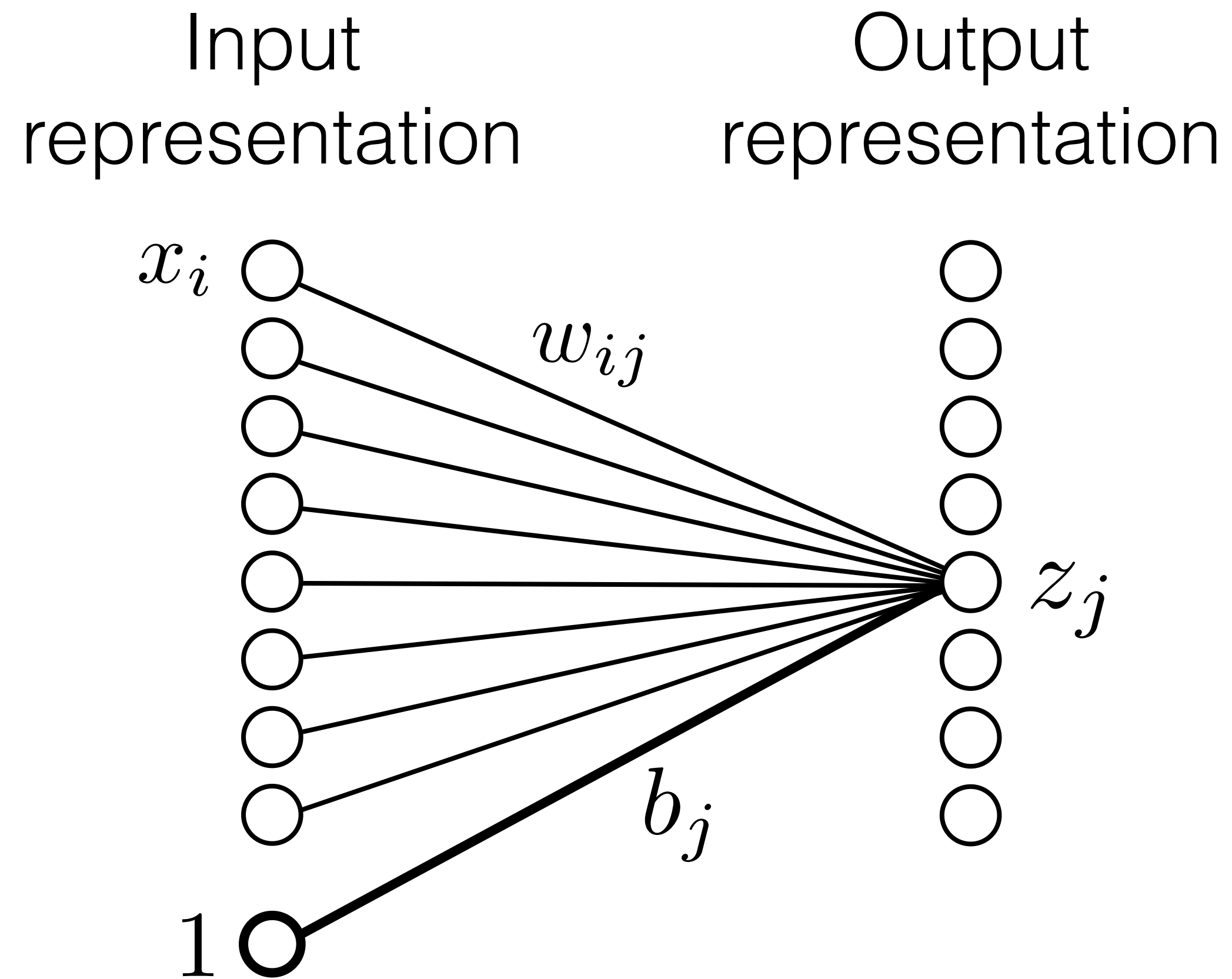
Linear layer



$$z_j = \sum_i w_{ij} x_i$$

Computation in a neural net

Linear layer



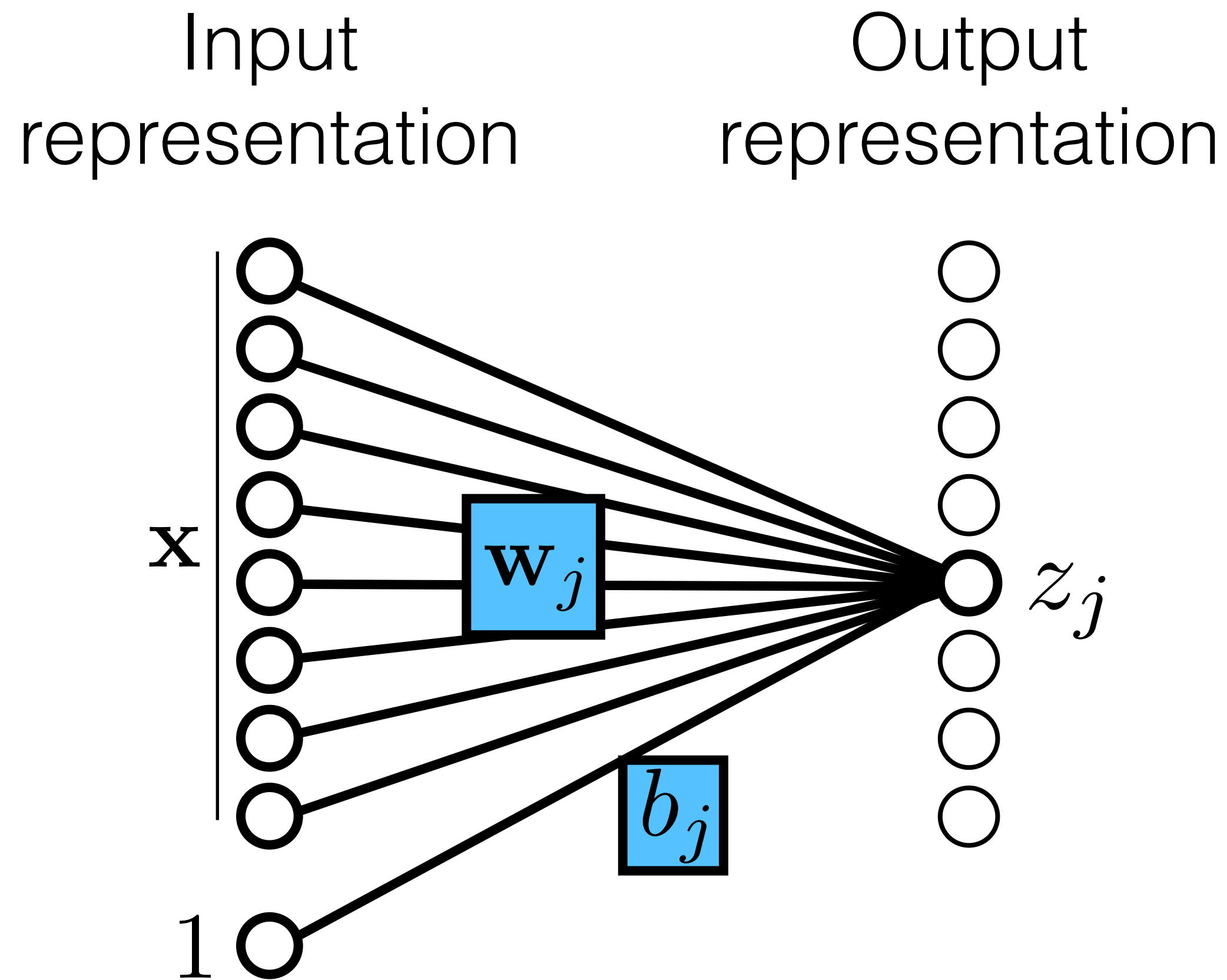
$$z_j = \sum_i w_{ij} x_i + b_j$$

weights

bias

Computation in a neural net

Linear layer



$$z_j = \mathbf{x}^T \mathbf{w}_j + b_j$$

weights

bias

$$\theta = \{\mathbf{W}, \mathbf{b}\}$$

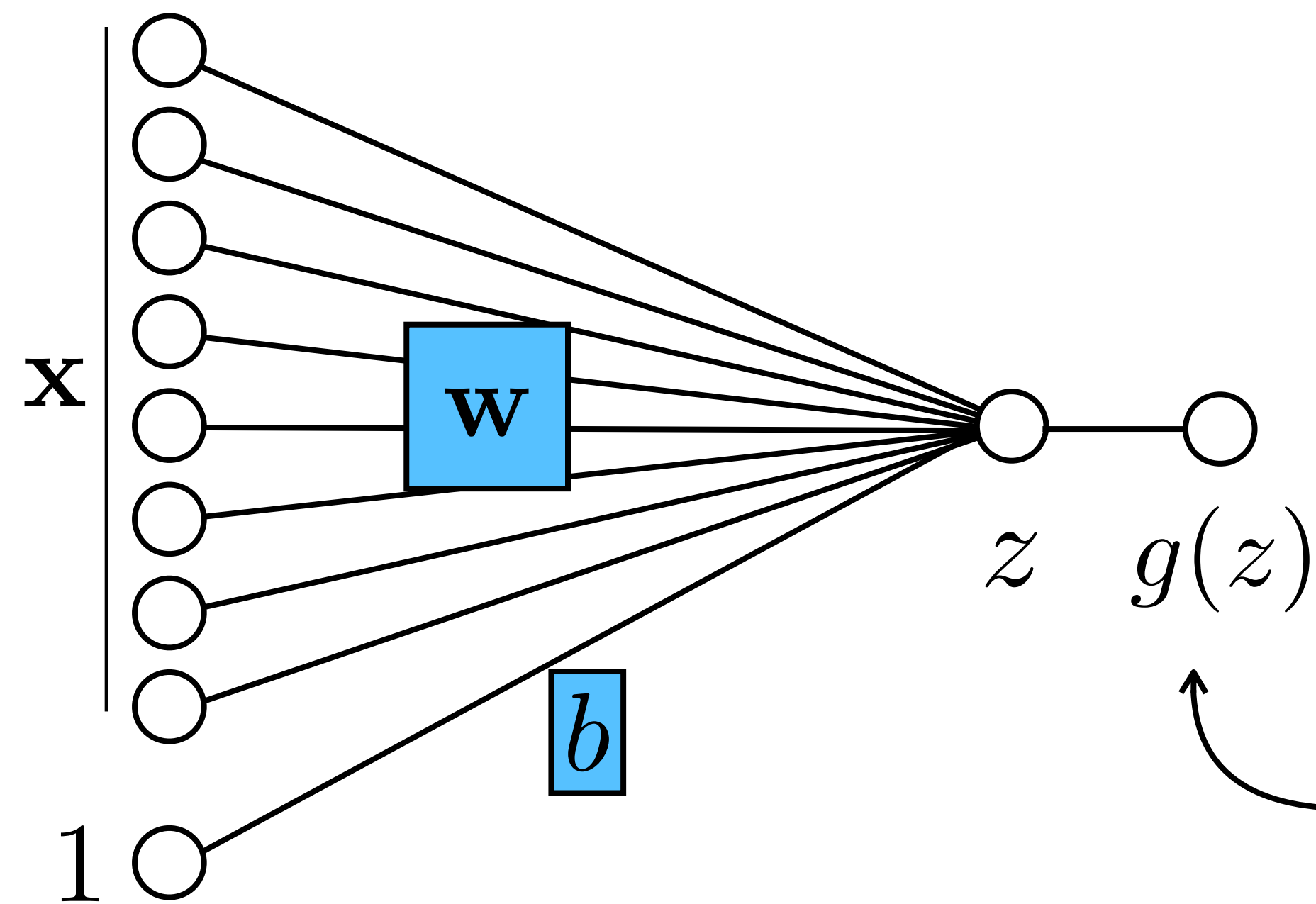
parameters of the model

Computation in a neural net — nonlinearity

“Perceptron”

Input
representation

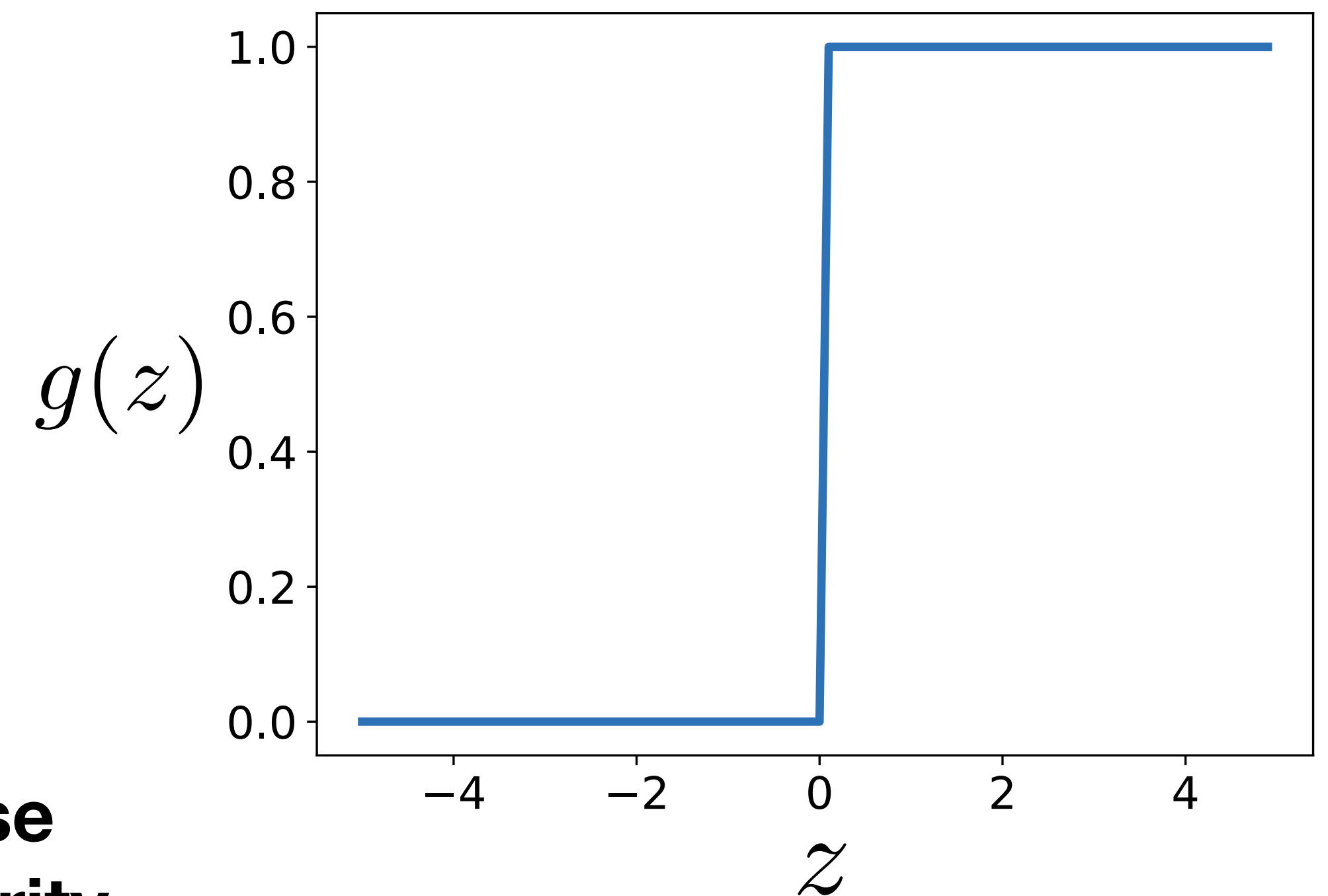
Output
representation



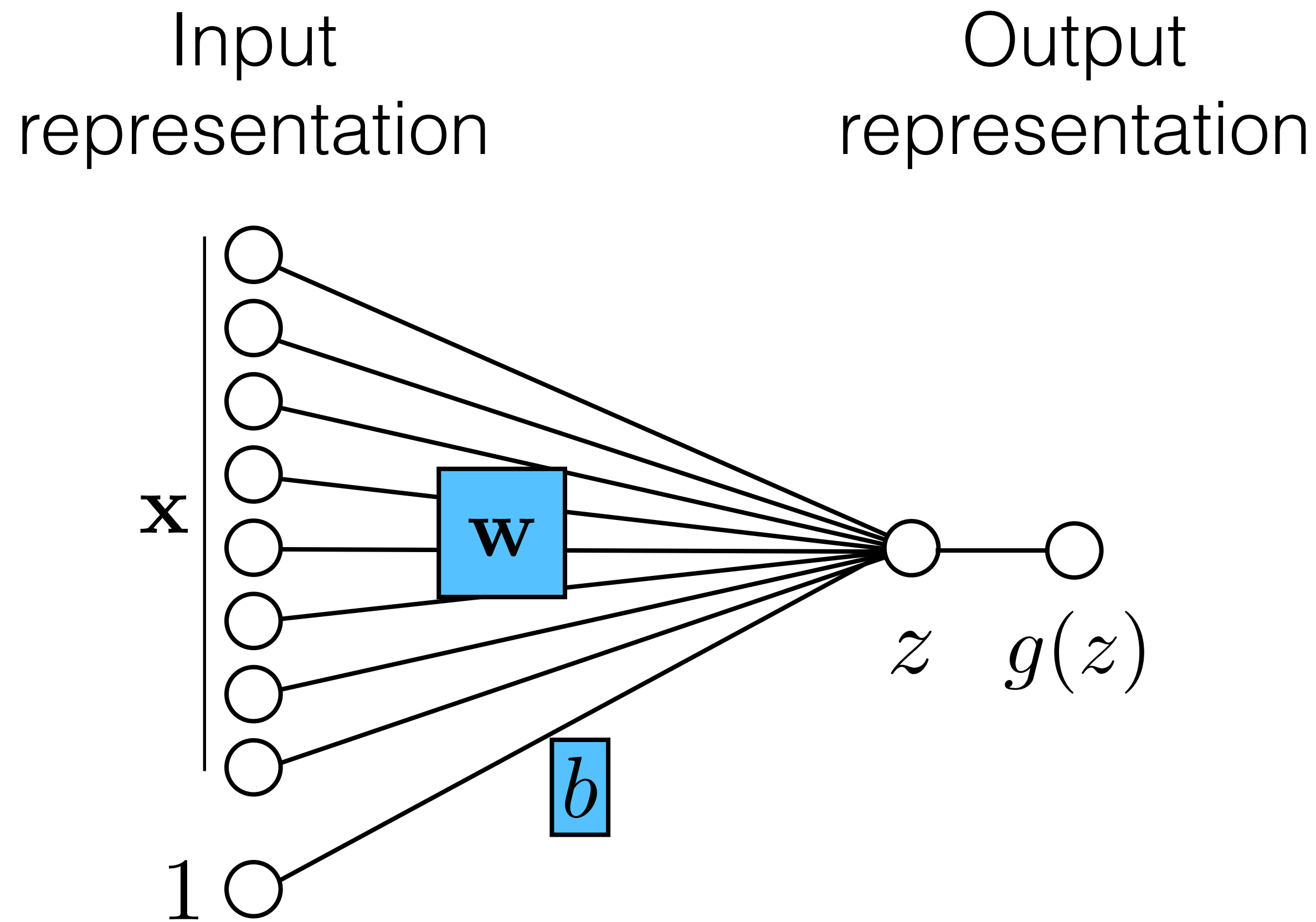
**Pointwise
Non-linearity**

Step function (“switch”)

$$g(z) = \begin{cases} 1, & \text{if } z > 0 \\ 0, & \text{otherwise} \end{cases}$$

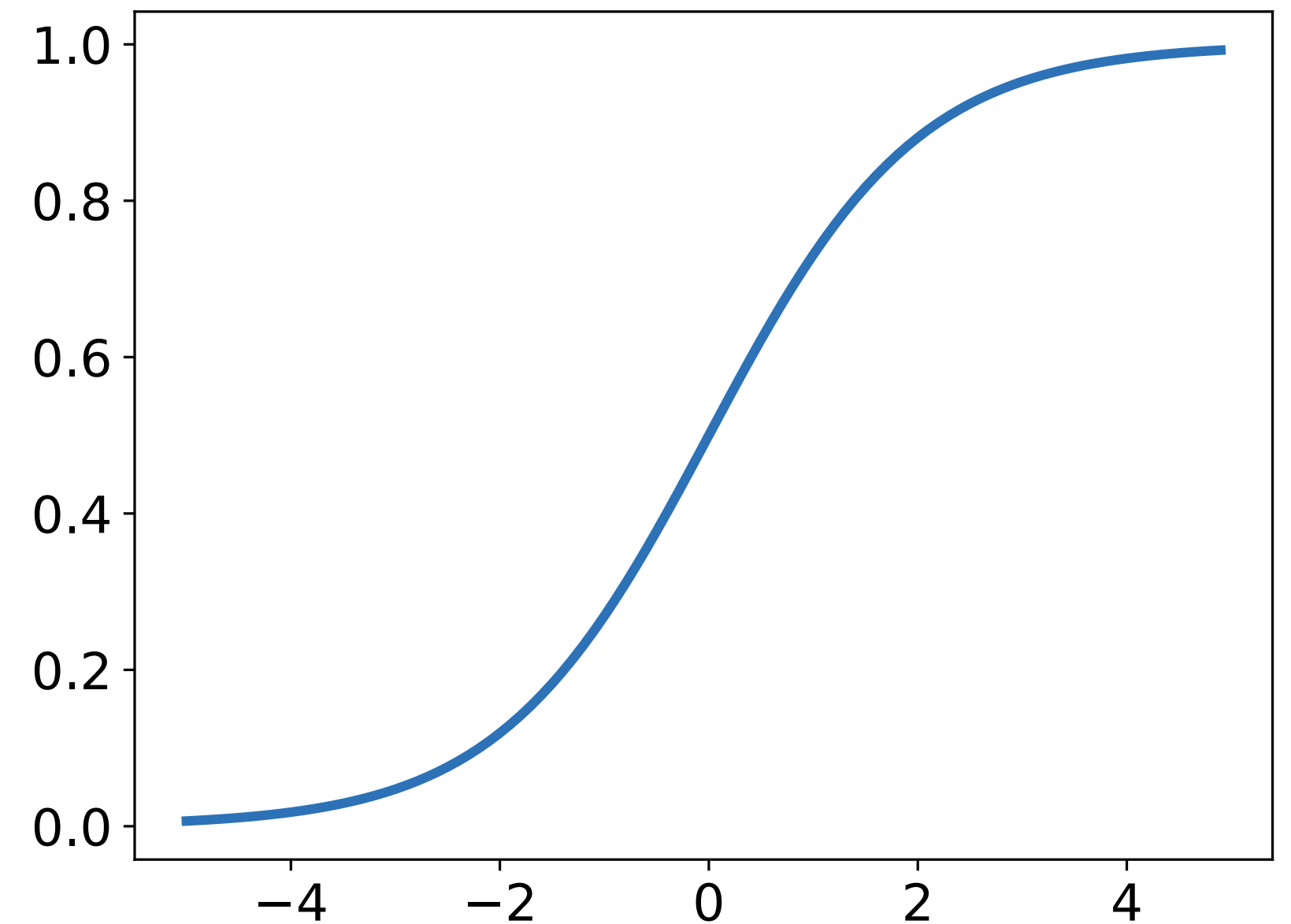


Computation in a neural net — nonlinearity



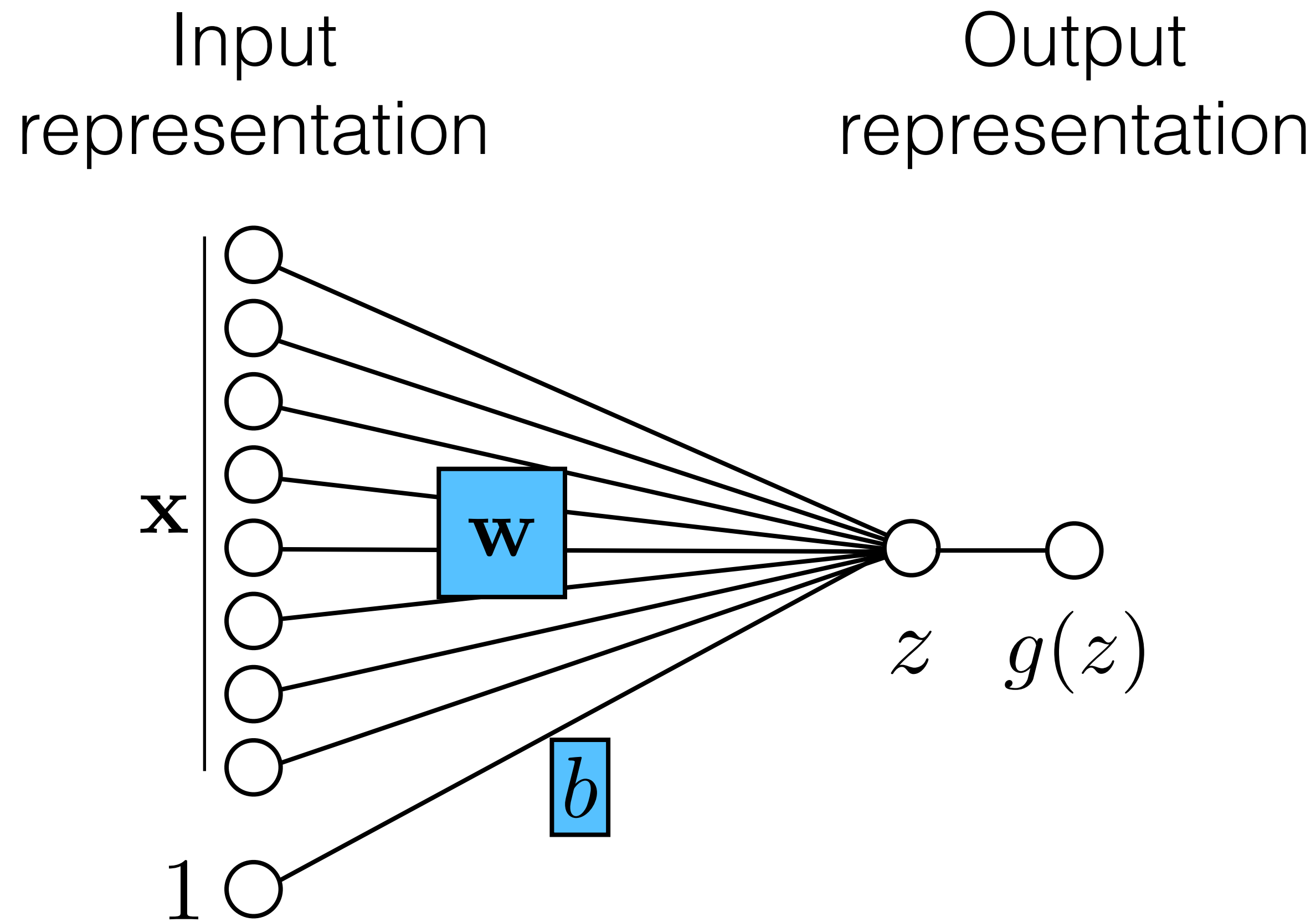
Sigmoid

$$g(z) = \frac{1}{1 + e^{-z}}$$



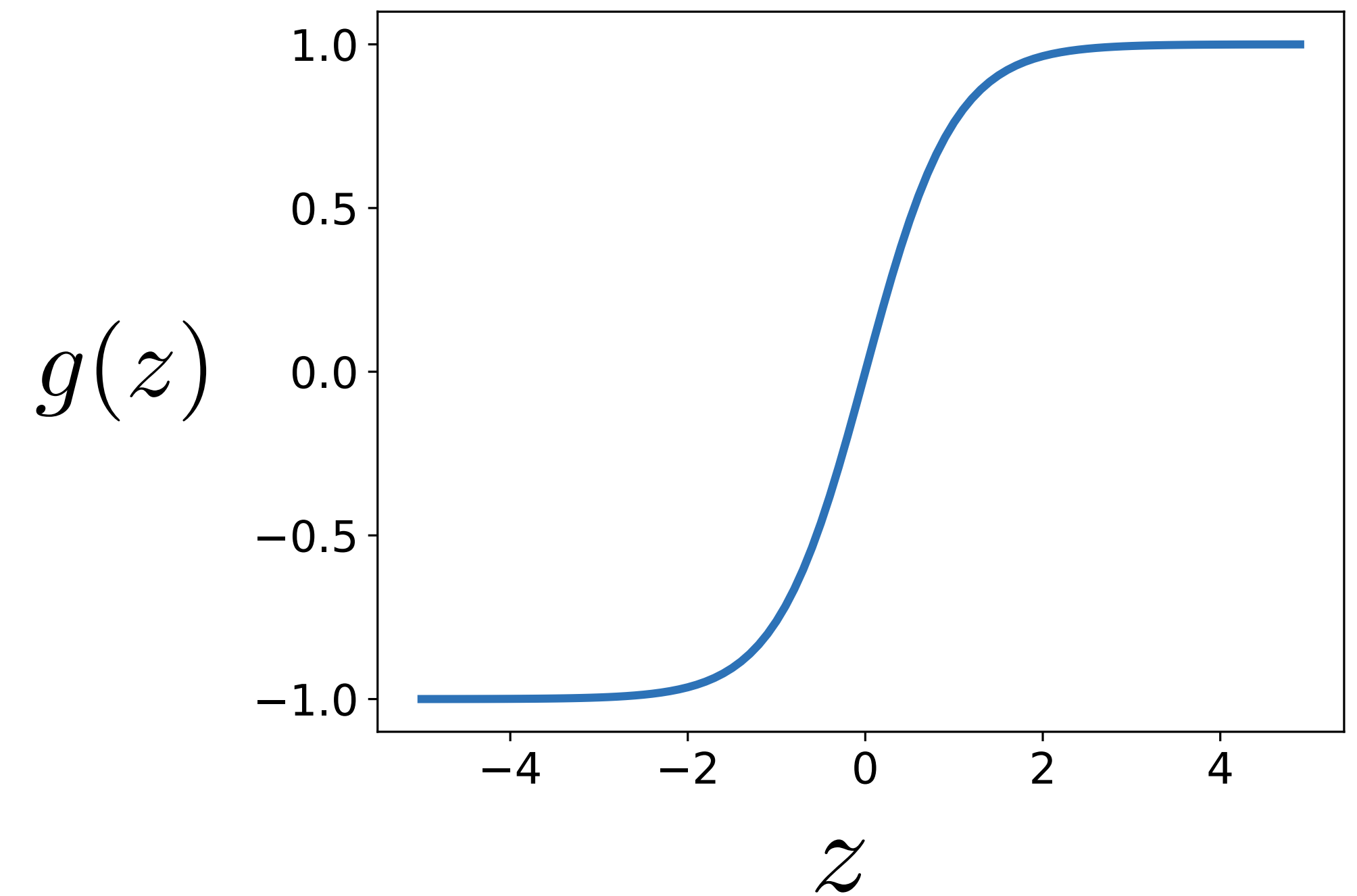
(soft step function)

Computation in a neural net — nonlinearity



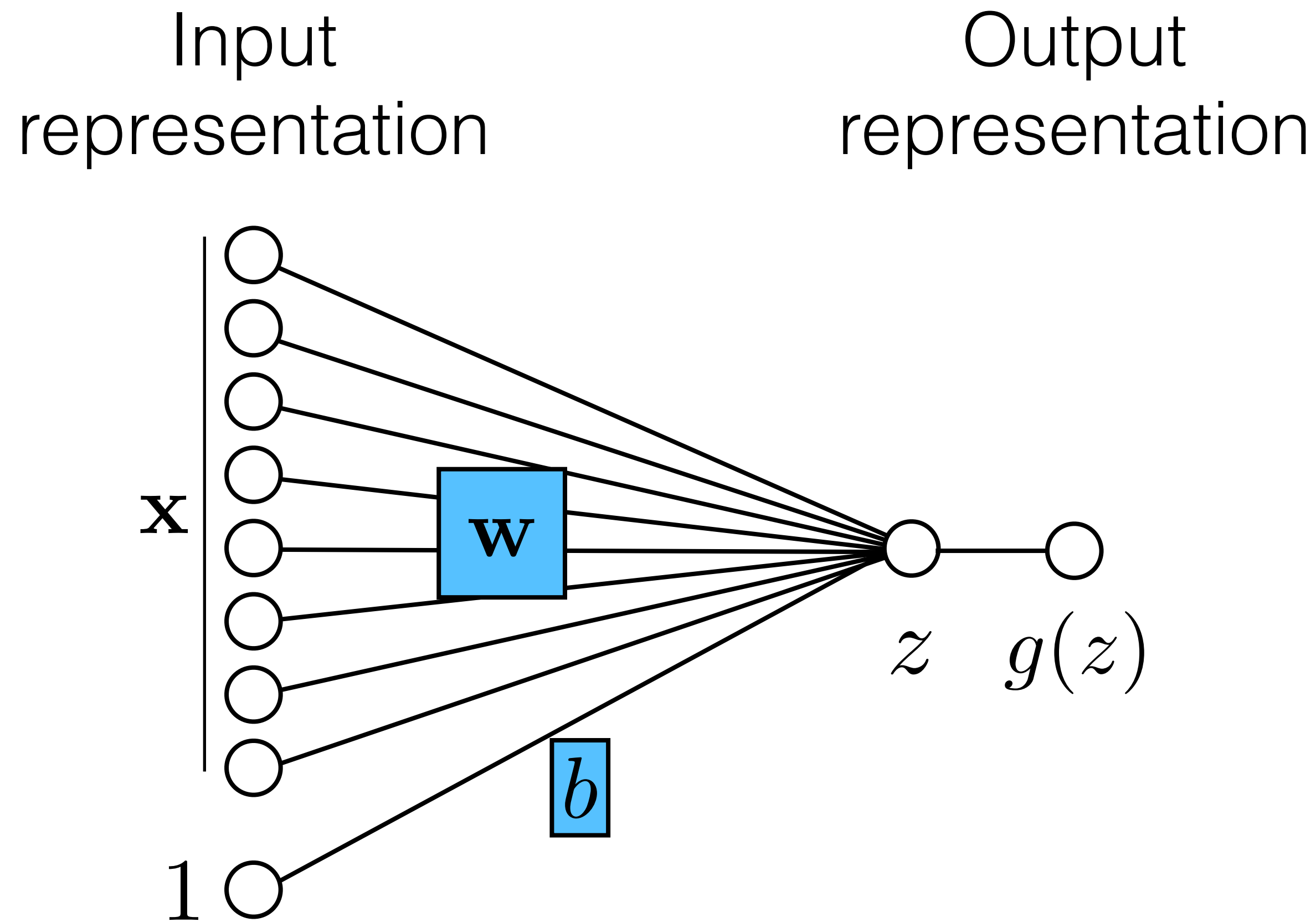
Tanh

$$g(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$



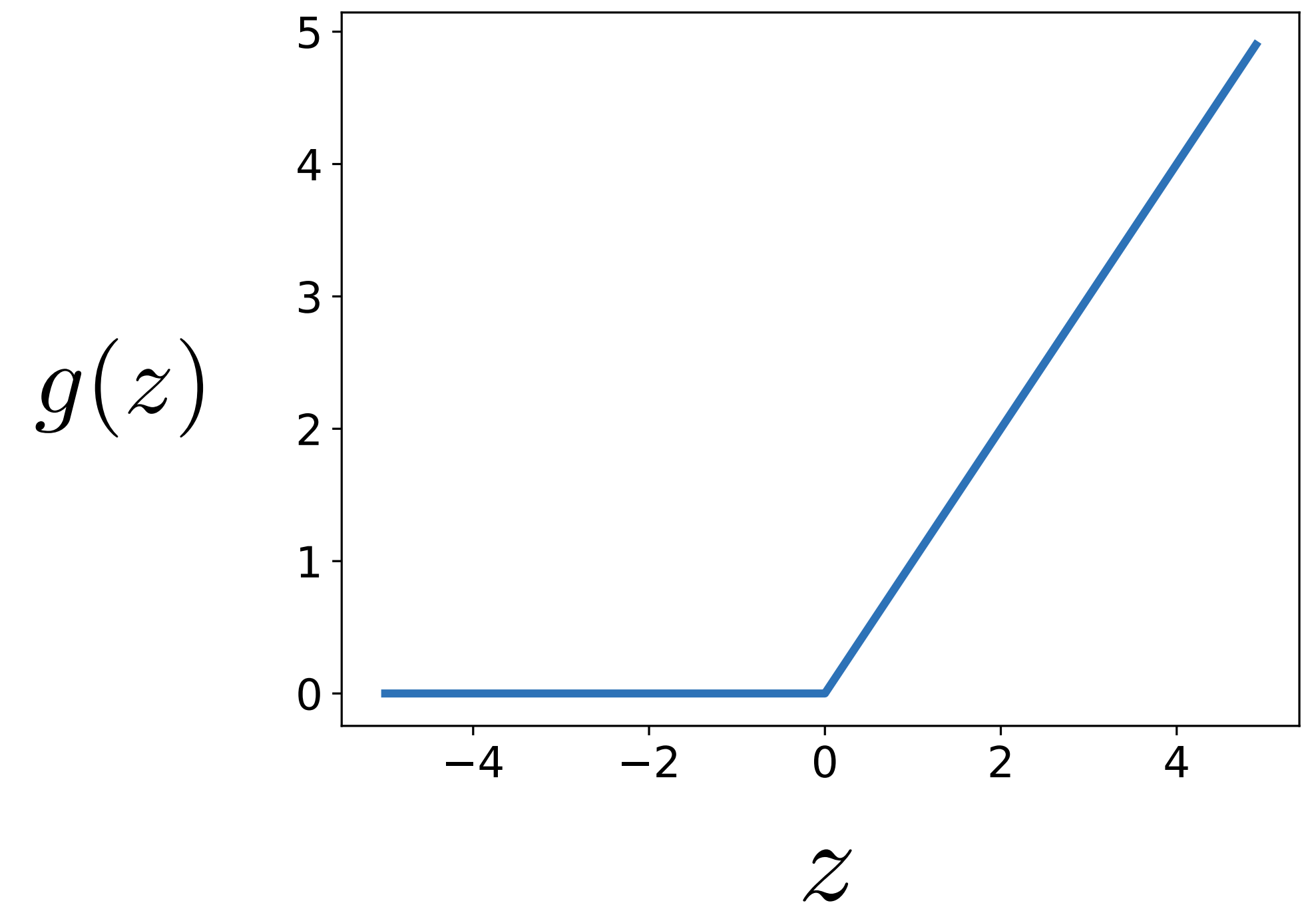
(soft, centered step function)

Computation in a neural net — nonlinearity



Rectified linear unit (ReLU)

$$g(z) = \max(0, z)$$



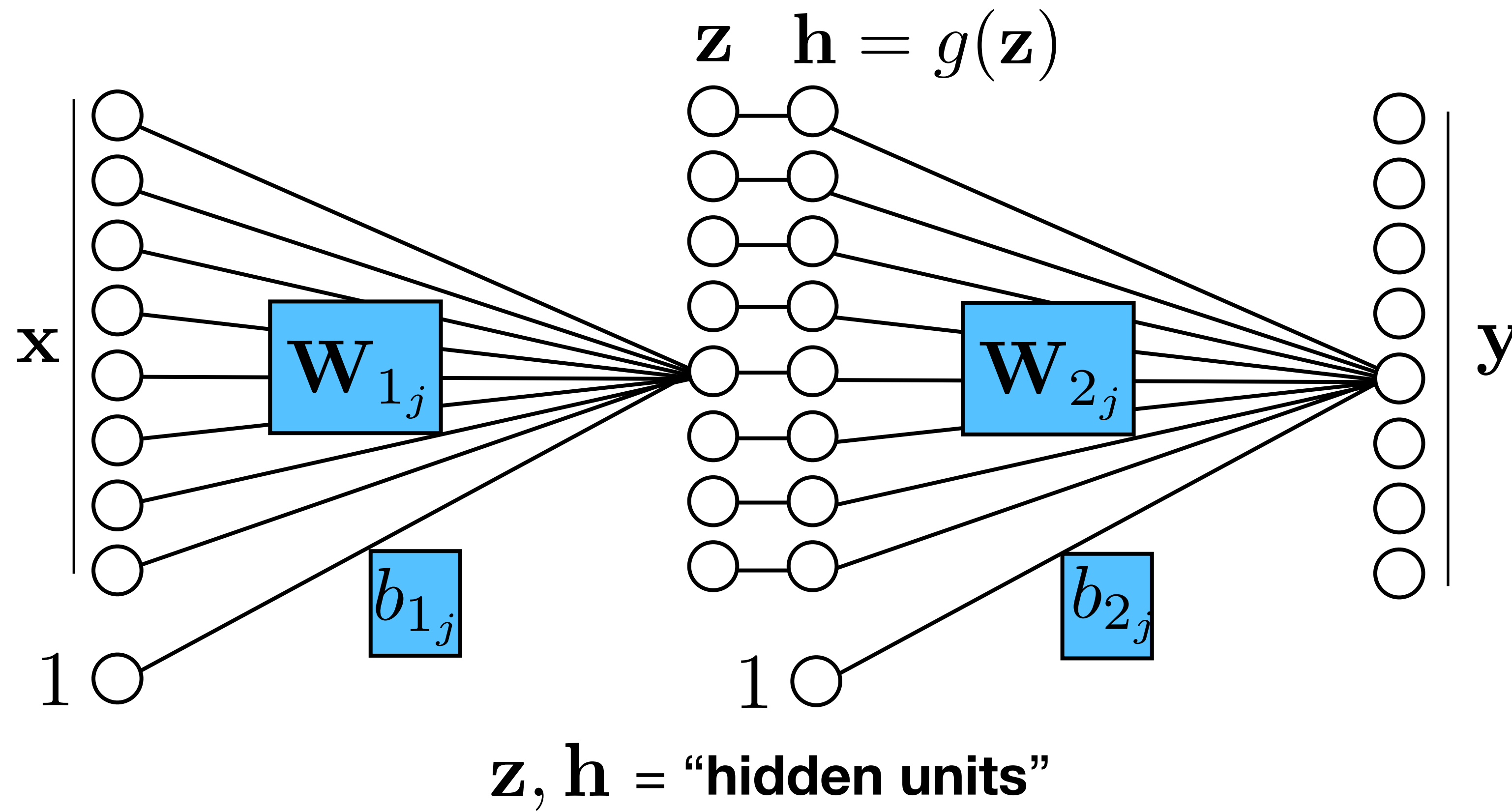
(core component in AlexNet,
“RevoLUtion of DL”)

Composing layers

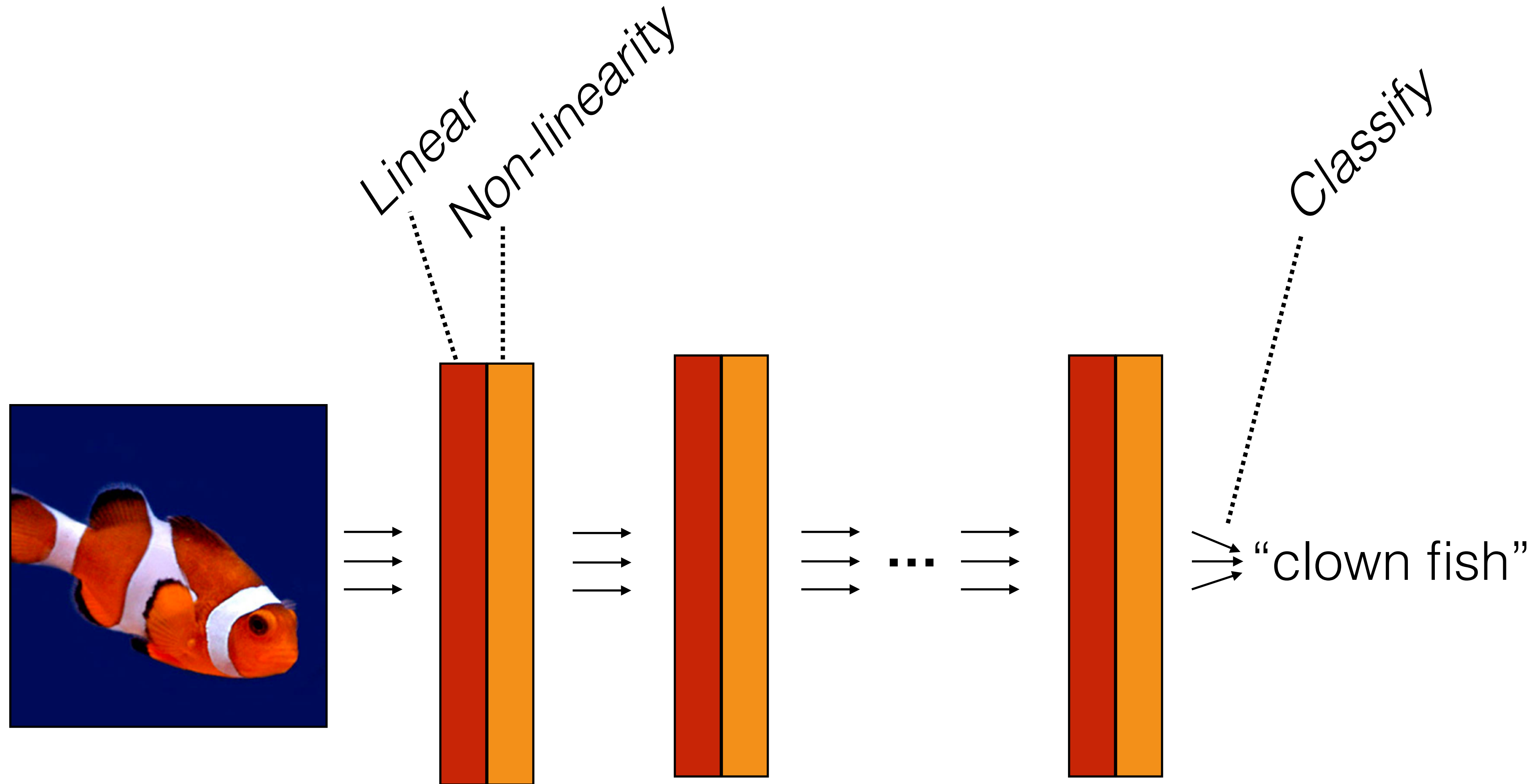
Input
representation

Intermediate
representation

Output
representation



Deep Neural Network

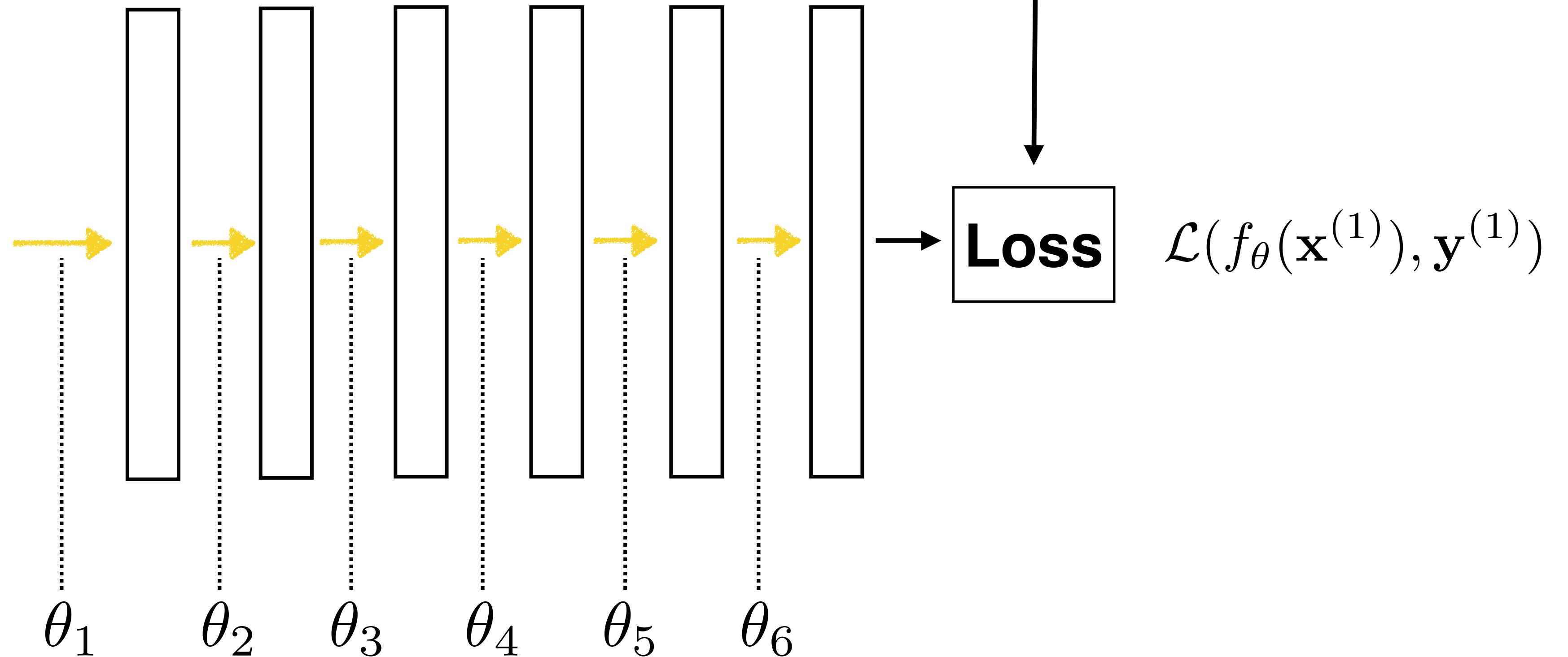
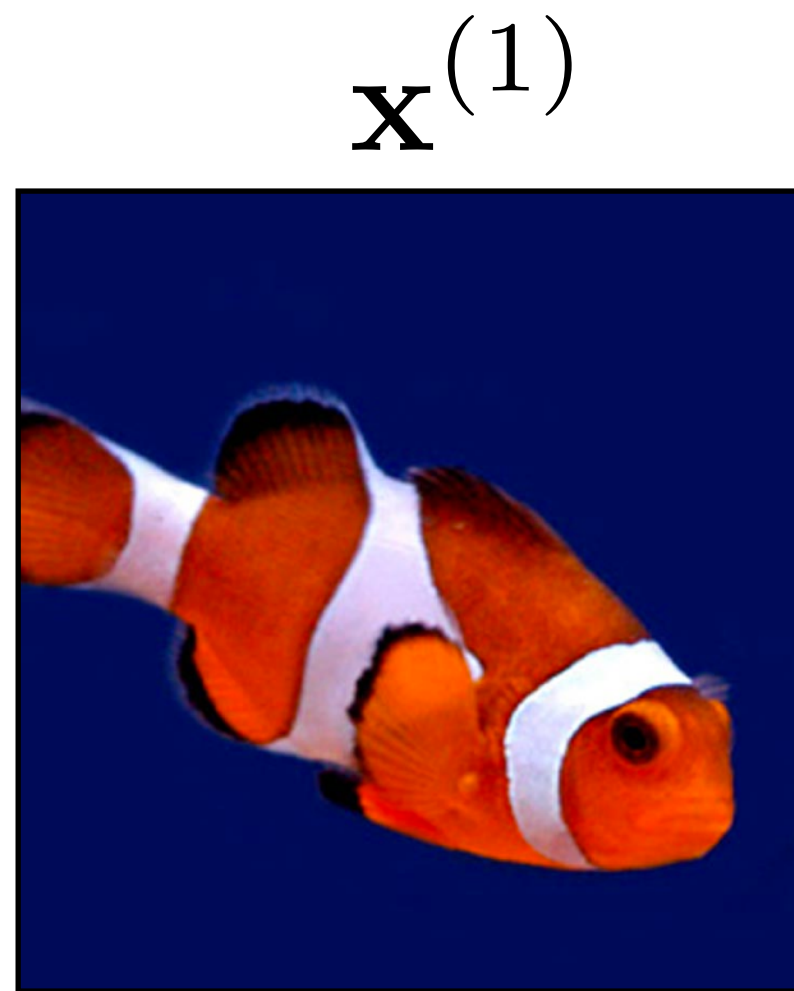


$$f(\mathbf{x}) = f_L(f_{L-1}(\dots f_2(f_1(\mathbf{x}))))$$

Deep Learning

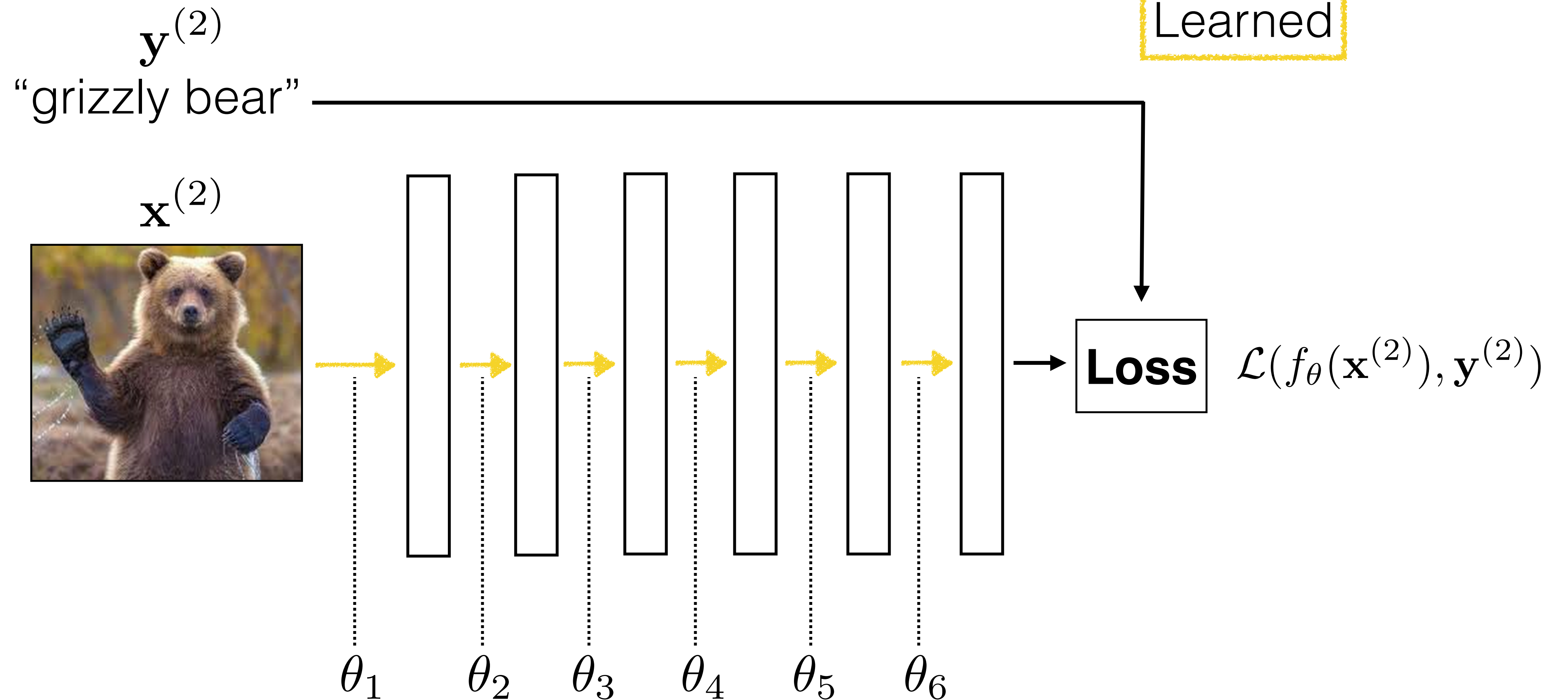
$\mathbf{y}^{(1)}$
“clown fish”

Learned



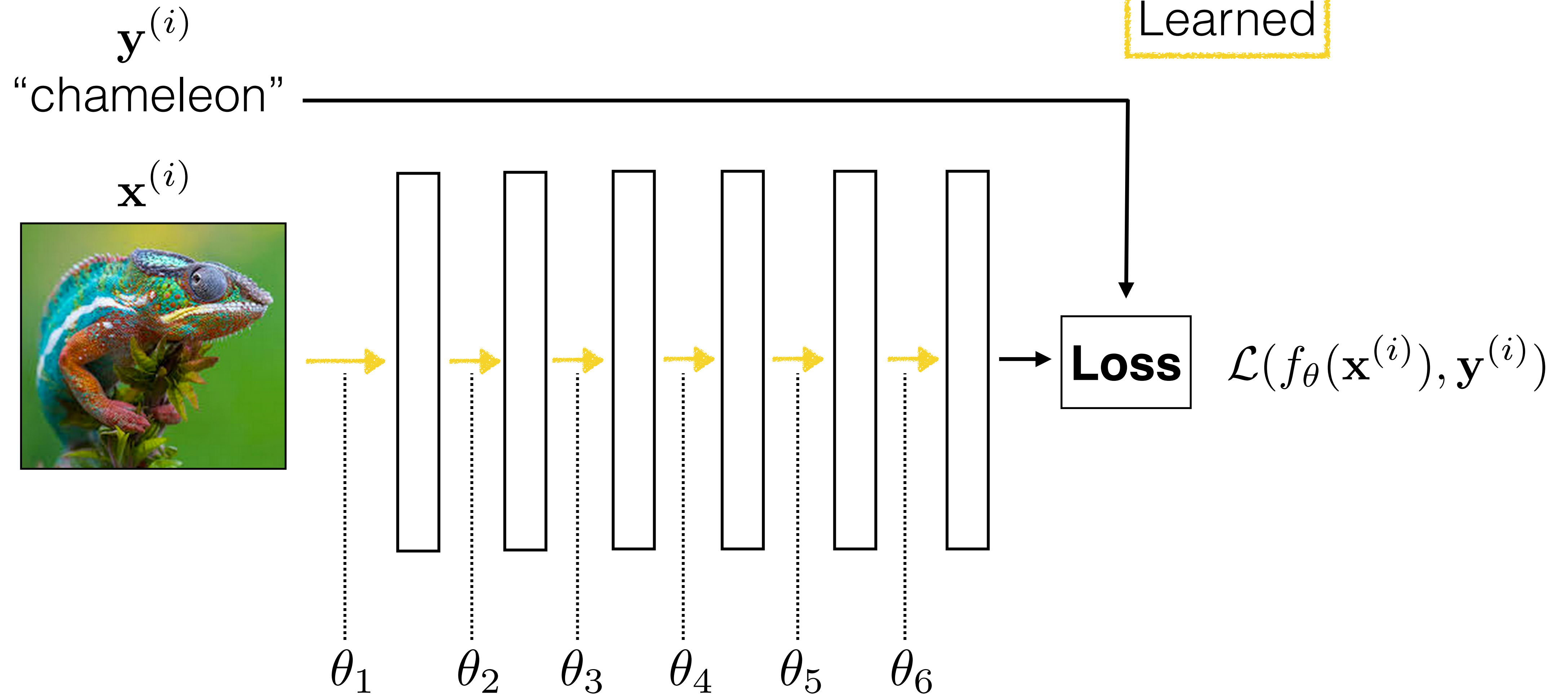
$$\theta^* = \arg \min_{\theta} \sum_{i=1}^N \mathcal{L}(f_{\theta}(\mathbf{x}^{(i)}), \mathbf{y}^{(i)})$$

Deep Learning



$$\theta^* = \arg \min_{\theta} \sum_{i=1}^N \mathcal{L}(f_{\theta}(\mathbf{x}^{(i)}), \mathbf{y}^{(i)})$$

Deep Learning



$$\theta^* = \arg \min_{\theta} \sum_{i=1}^N \mathcal{L}(f_{\theta}(\mathbf{x}^{(i)}), \mathbf{y}^{(i)})$$

Training by Gradient descent

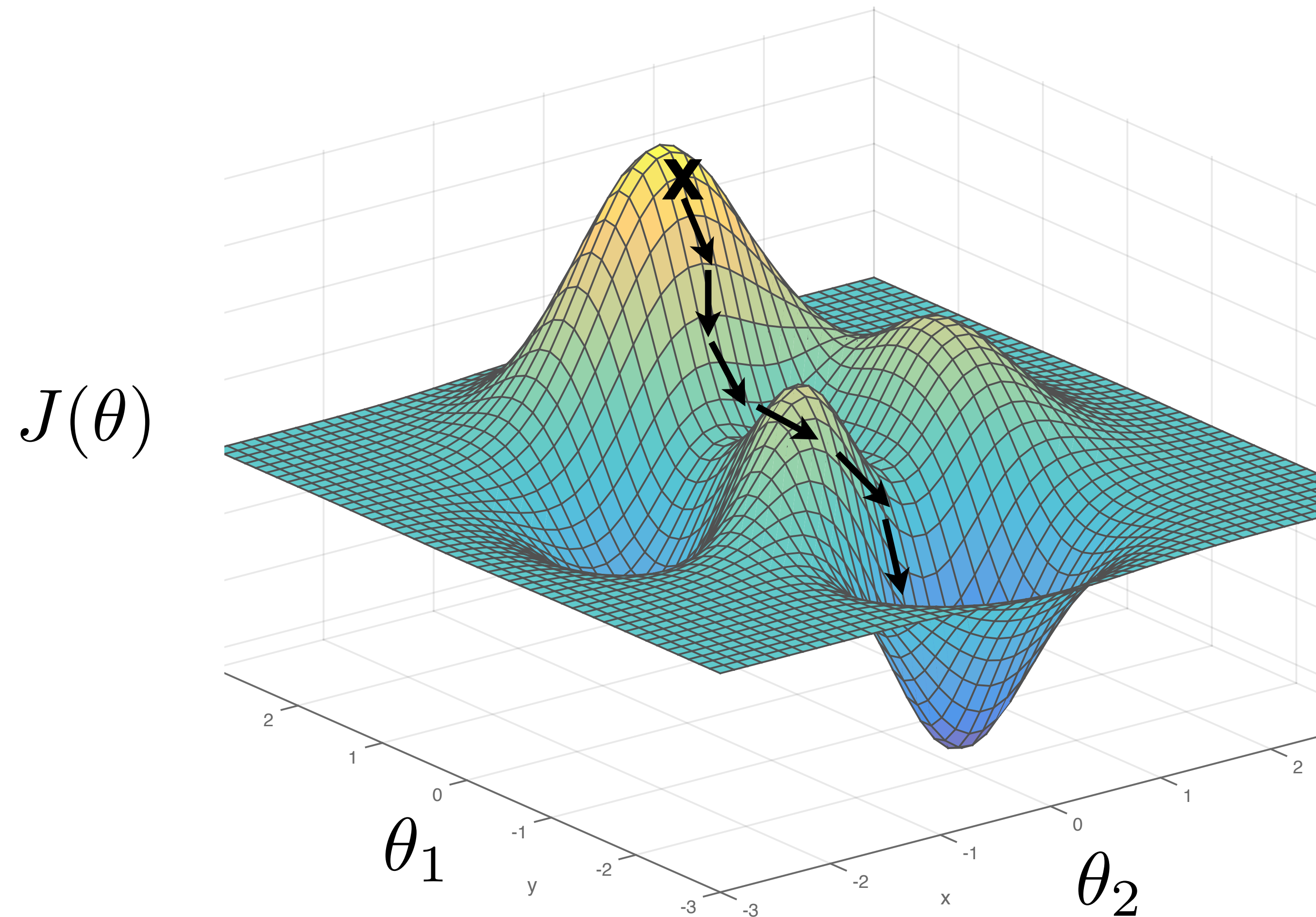
$$\theta^* = \arg \min_{\theta} \underbrace{\sum_{i=1}^N \mathcal{L}(f_{\theta}(\mathbf{x}^{(i)}), \mathbf{y}^{(i)})}_{J(\theta)}$$

One iteration of gradient descent:

$$\theta^{t+1} = \theta^t - \eta_t \left. \frac{\partial J(\theta)}{\partial \theta} \right|_{\theta=\theta^t}$$

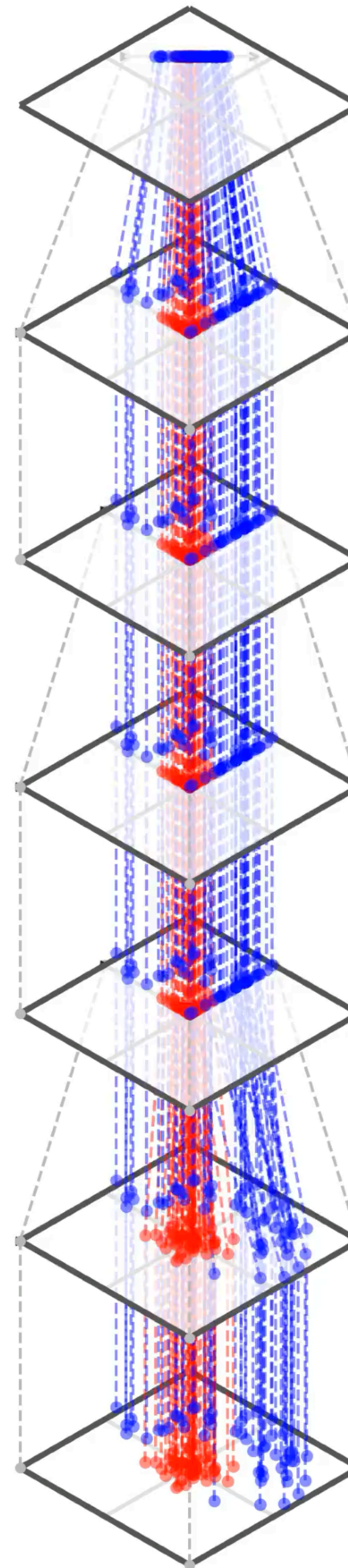
learning rate

Gradient descent



$$\theta^* = \arg \min_{\theta} J(\theta)$$

Loss: 0.65



softmax

linear

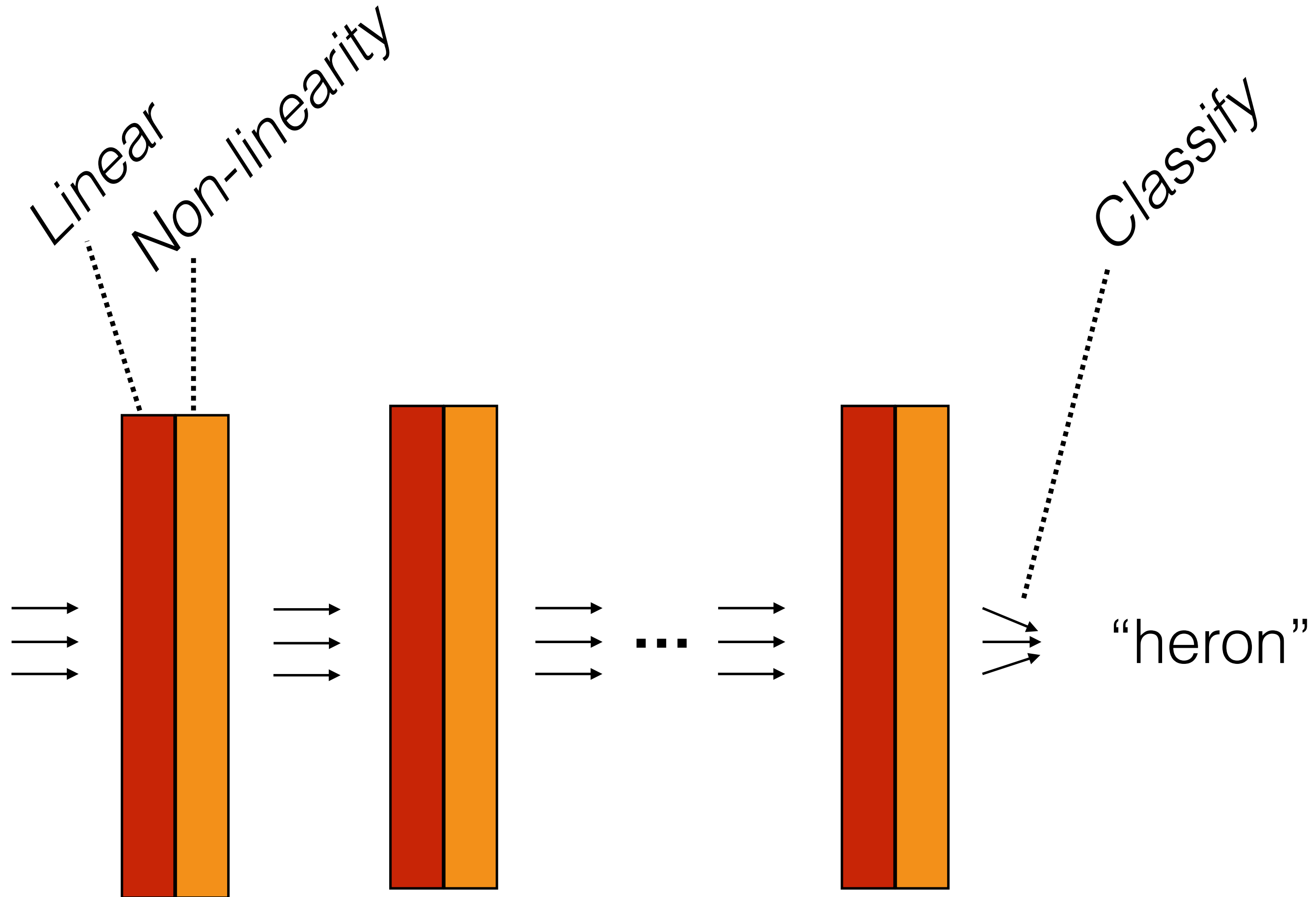
relu

linear

relu

linear

Inference



$$\theta^* = \arg \min_{\theta} J(\theta)$$

Modern Deep Neural Networks follow this general framework.

But:

- What are the building blocks?
- How do we optimize the objective functions?
- How do we learn useful representations and models?
- How do we apply them to real-world problems?

This course will cover these questions.

1. Introduction to Deep Learning

- A Brief History of (Artificial) Neural Networks
- A Quick Look at Deep Neural Networks